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PHYSICS

1) A spherical shell is uniformly charged by a charge q. A point charge q is at its center. The work of electric force upon the expansion of shell from R to 2R is :-

(1) $\frac{3q^2}{16\pi\varepsilon_0 R}$

(2) $\frac{q^2}{8\pi\varepsilon_0 R}$

(3) $\frac{q^2}{16\pi\varepsilon_0 R}$

(4) $\frac{q^2}{4\pi\varepsilon_0 R}$

2) A non conducting sphere of radius 'a' has a net charge '+q' uniformly distributed throughout its volume. A hollow spherical conductor having inner and outer, radii 'b' and 'c' and net charge '–q' is



concentric with the sphere (see the figure) Read the following statements (i) The electric field at a distance r from the centre of the sphere = $\frac{1}{4\pi\varepsilon_0}\frac{qr}{a^3}$ for r < a
(ii) The electric field at distance r from the centre of the sphere for a < r < b = 0
(iii) The electric field at distance r from the centre of the sphere for b < r < c = 0
(iv) The charge on the inner surface of the hollow sphere = –q
(v) The charge on the outer surface of the hollow sphere = +q
Which of the above statements are true ?

(1) (i) , (ii) and (v)

(2) (i) , (iii) and (iv)

(3) (ii) , (iii) and (iv)

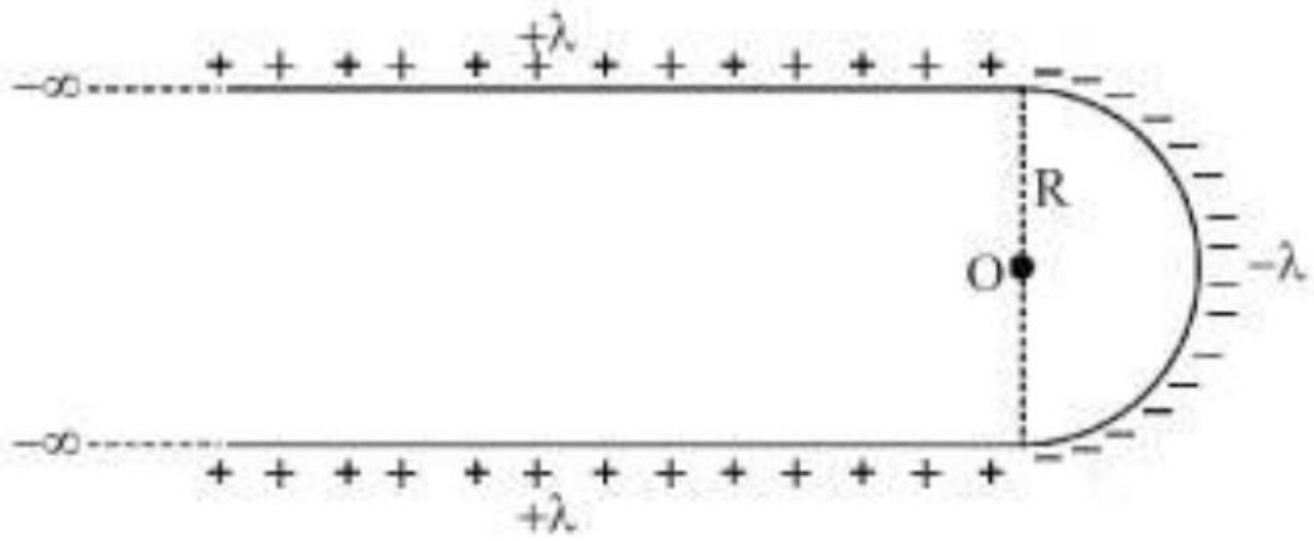
(4) (ii) , (iii) and (v)

3) An infinitely long wire is bent at the middle in form of a semicircular bend. Uniform linear charges are distributed as shown in figure. Then net electric field at point O is :

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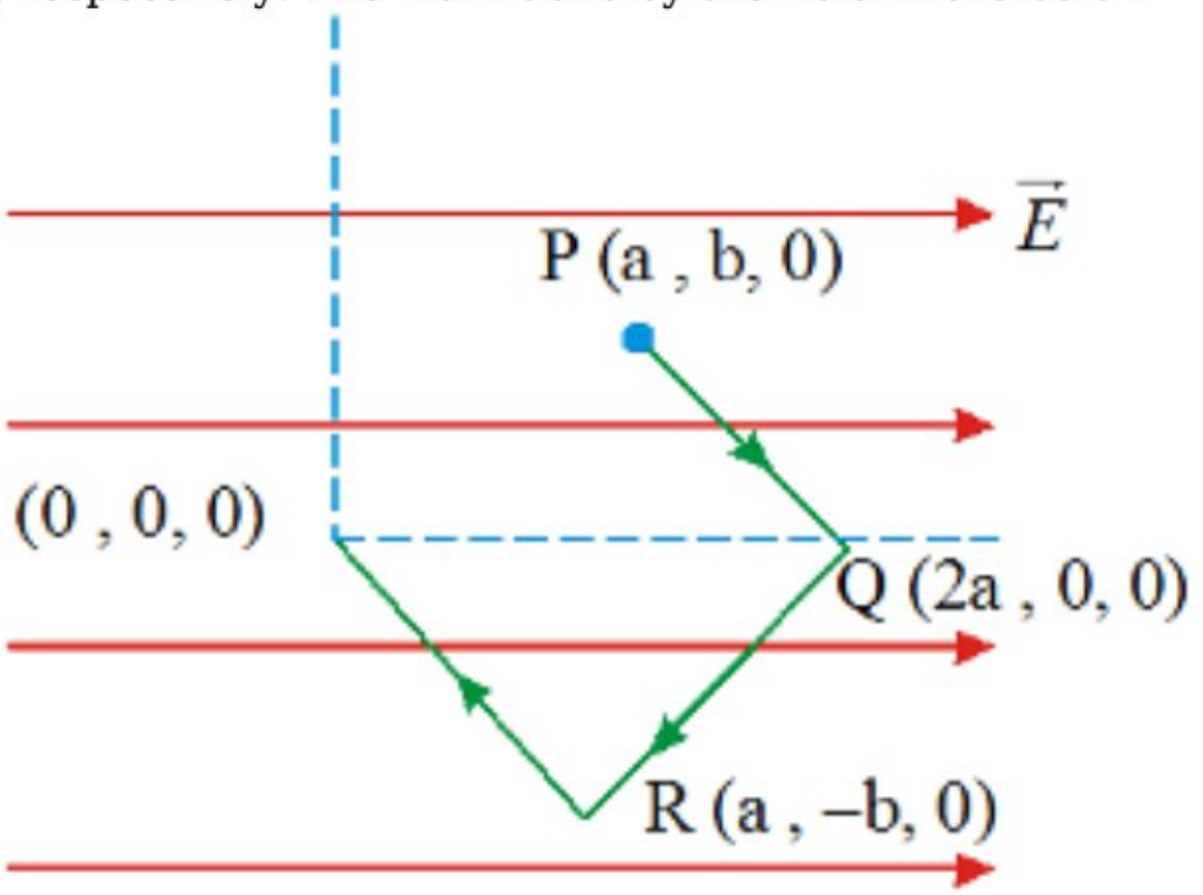
(1) $\frac{K\lambda}{R}$

(2) $\frac{2K\lambda}{R^2}$

(3) $\frac{4K\lambda}{R}$

(4) $\frac{8K\lambda}{2}$

4) A point charge q moves from point P to point S along the path PQRS in a uniform electric field $\vec{E}$ pointing parallel to the positive direction of the x-axis. The coordinates of the points P, Q, R and S are (a,b,0), (2a,0,0), (a, –b, 0) and (0,0,0) respectively. The work done by the field in the below



process is given by the expression

(1) qaE

(2) –qaE

(3) $q(\sqrt{a^2+b^2})E$

(4) $3qE\sqrt{a^2+b^2}$

5) A point charge q having mass m is projected from a long distance with speed v towards another stationary particle of same mass and charge. The distance of closest approach of ther particles is

(1) $\frac{q^2}{2\pi\varepsilon_0 mv^2}$

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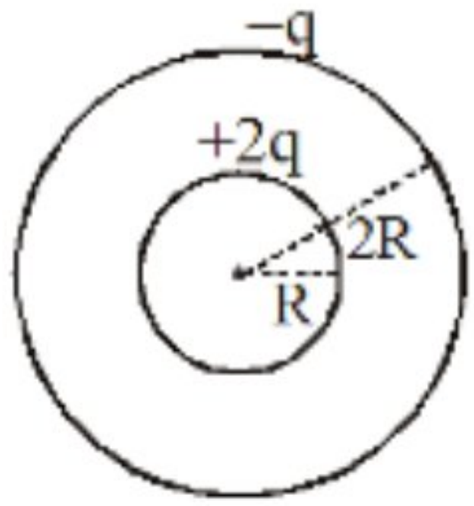
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(2) $\frac{2q^2}{\pi\varepsilon_0 mv^2}$

(3) $\frac{q^2}{\pi\varepsilon_0 mv^2}$

(4) Zero



6) Charge flow into earth when outer sphere is connected to earth

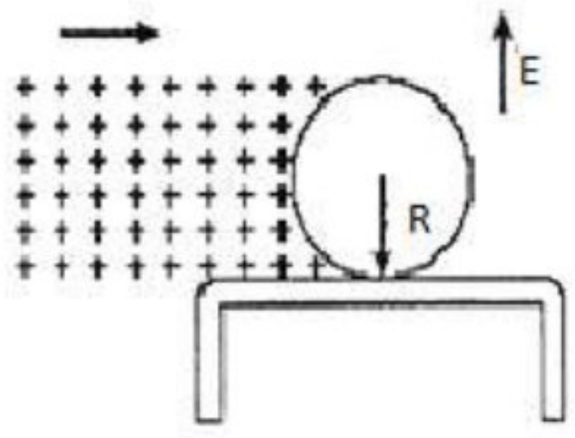
(1) +q

(2) –q

(3) +2q

(4) –2q

7) Figure shows a neutral metal ball of mass 50 kg and radius $\frac{2}{\sqrt{\pi}}$m placed on an insulating horizontal surface. An upward electric field of strength 5 x 10^5 N/C exists in the space. A stream of light ions is switched on at t = 0. The ions are incident on the ball from left side at a speed 2 x 10^6 m/s. The charge density of ion beam is
5 x 10^{-12} C/m^3. All charges incident on the ball are absorbed. Friction is large enough to prevent any horizontal movement of the ball. Neglect any effect of the electric field on the ion beam. The time in second at which ball will rise up from the surface is



(1) 25 s

(2) 15 s

(3) 20 s

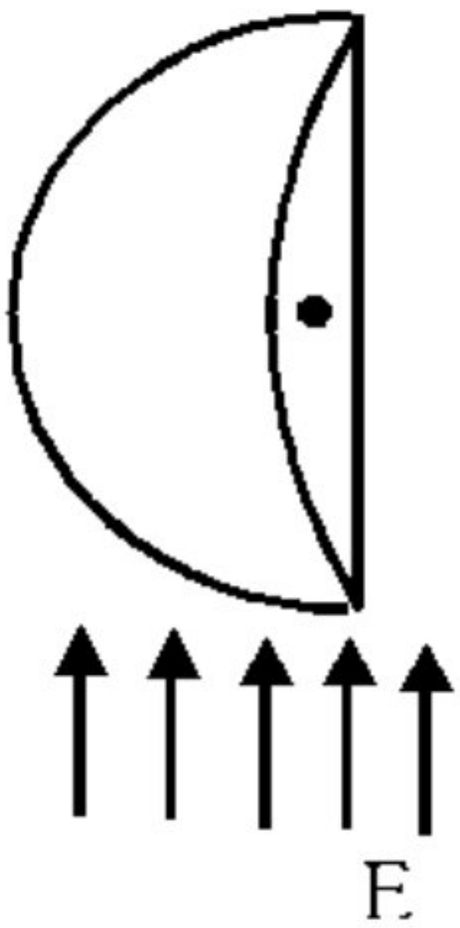
(4) 5 s

8) A hemisphere of radius R is kept in an uniform electric field as shown in figure. The flux entering

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in the sphere is :-

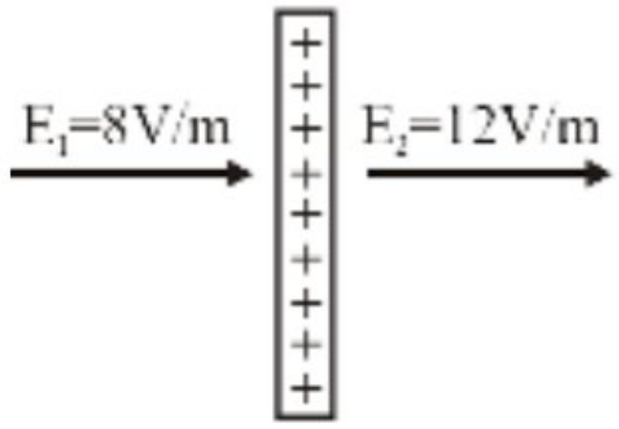
(1) $\frac{\pi R^2}{2}E$

(2) pR^2E

(3) Zero

(4) $2\pi R^2E$

9) The net electric field on two sides of a large charged plate kept in an external electric field is shown in the figure. The charge density on the plate in S.I. units is given by (ε_0 is the Permittivity



of free space in S.I. units)

(1) $2\varepsilon_0$

(2) $4\varepsilon_0$

(3) $10\ \varepsilon_0$

(4) 0

10) There are four concentric conducting shells A, B, C and D of radii a, 2a, 3a and 4a respectively. Shells B and D are given charges +q and –q respectively. Shell C is now earthed. The potential difference $V_A - V_C$ is $k = \frac{1}{4\pi\varepsilon_0}$

(1) $\frac{kq}{2a}$

(2) $\frac{kq}{3a}$

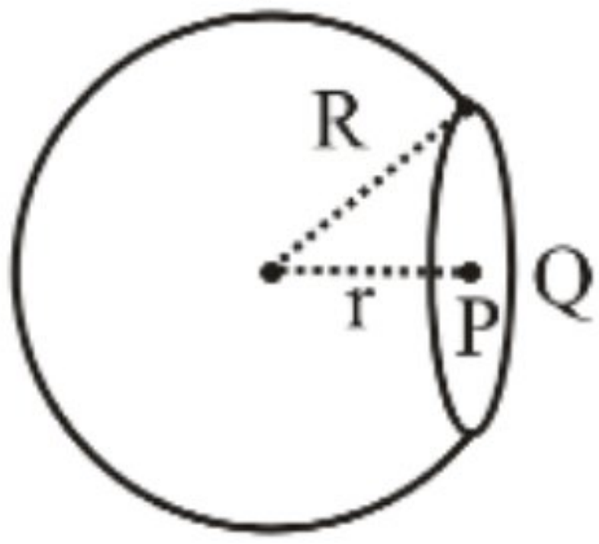
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(3) $\frac{kq}{4a}$

(4) $\frac{kq}{6a}$

11) Point charge Q is placed at point P in the plane of circular face of sphere. The electric flux



passing through the spherical part shown in the figure is:-

(1) $\frac{Q}{\epsilon_0}$

(2) $\frac{Q}{2\epsilon_0}$

(3) $\frac{Q}{3\epsilon_0}$

(4) $\frac{Q}{2\epsilon_0}\left(1+\frac{r}{R}\right)$

12) A charge water drop of radius "r" is in equilibrium in an electric field. If charge on it is equal to charge of an electron, then intensity of electric field will be : (Density of water = ρ, assume gravity)

(1) $\frac{8\pi r^3\rho}{3e}g$

(2) $\frac{\pi r^3\rho}{3e}g$

(3) $\frac{6\pi r^3\rho}{15e}g$

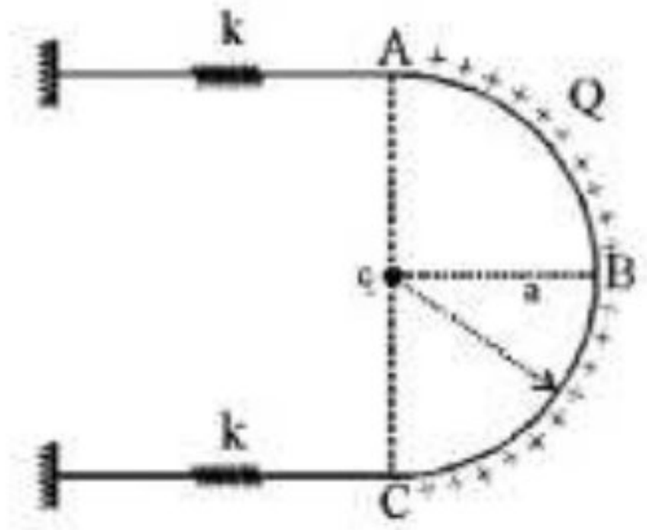
(4) $\frac{4\pi r^3\rho}{3e}g$

13) In the given figure, ABC is a nonconducting semicircular wire of radius a carrying total charge Q uniformly distributed on it and a point charge q is at its centre. Ends of the wire are attached to, two separate springs, each having spring constant k as shown in the figure. In the given position, system is in equilibrium. Now the point charge q is suddenly removed. The amplitude of resulting oscillation

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is (ignore any force other than spring force and electrostatic force)

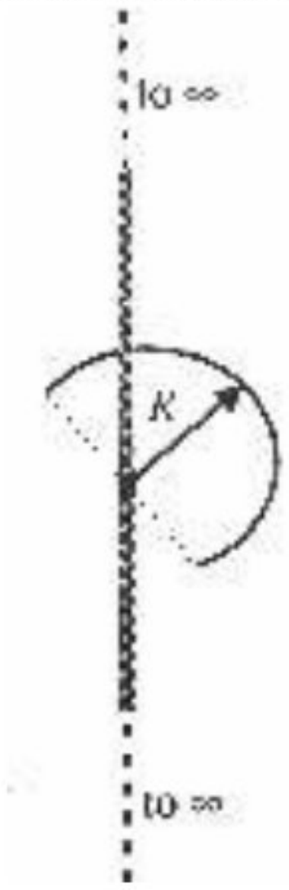
(1) $\frac{qQ}{8\pi^2\varepsilon_0 a^2 k}$

(2) $\frac{qQ}{4\pi^2\varepsilon_0 a^2 k}$

(3) $\frac{qQ}{8\pi\varepsilon_0 a^2 k}$

(4) $\frac{qQ}{4\pi\varepsilon_0 a^2 k}$

14) Find the force experienced by a semicircular rod having a charge q as shown in Fig. Radius of the wire is R, and the line of charge with linear charge density λ passes through its centre and is



perpendicular to the plane of wire.

(1) $\frac{\lambda q}{2\pi^2\in_0 R}$

(2) $\frac{\lambda q}{\pi^2\in_0 R}$

(3) $\frac{\lambda q}{4\pi^2\in_0 R}$

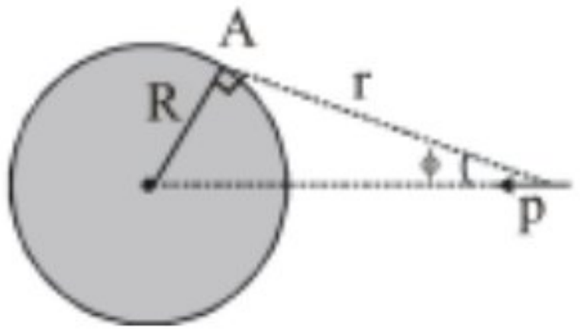
(4) $\frac{\lambda q}{4\pi\in_0 R}$

15) A dipole having dipole moment p is placed in front of a solid uncharged conducting sphere as shown in the diagram. The net potential at point A lying on the surface of the sphere is :-

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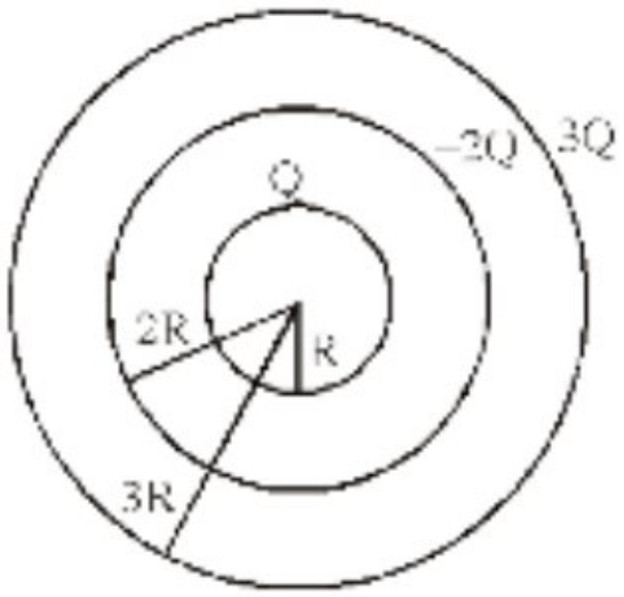
(1) $\frac{kp\cos\phi}{r^2}$

(2) $\frac{kpcos^2\phi}{r^2}$

(3) 0

(4) $\frac{2kpcos^2\phi}{r^2}$

16) Three meltotic shells of radius R, 2R and 3R has charges Q, –2Q and 3Q respectively. If middle shell and outer most shell are connected by wire then what will be charge in middle shell.



(1) Zero

(2) Q

(3) −Q

(4) 3Q

17) If an electron enters into a space between the plates of a parallel plate capacitor at an angle α with the plates and leaves at an angle β to the plates, the ratio of its kinetic energy while entering the capacitor to that while leaving will be :

(1) $(\cos\alpha/\cos\beta)^2$

(2) $(\cos\beta/\cos\alpha)^2$

(3) $(\sin\alpha/\sin\beta)^2$

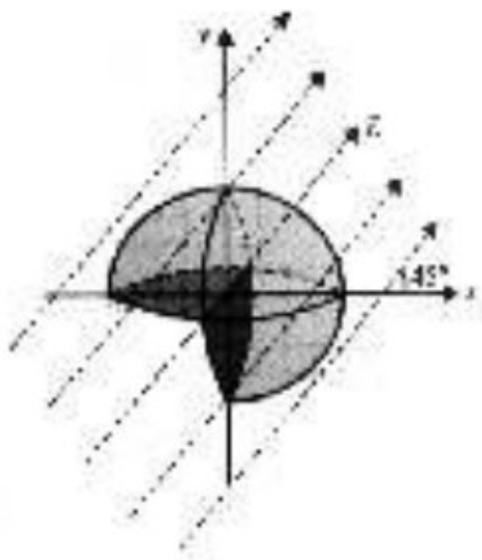
(4) $(\sin\beta/\sin\alpha)^2$

18) One-fourth of a sphere of radius R is removed as shown in figure. An electric field E exists

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parallel to the xy plane. Find the flux through the remaining curved part:

(1) $\pi R^2 E$

(2) $\sqrt{2}\pi R^2 E$

(3) $\pi R^2 E/\sqrt{2}$

(4) None of these

19) A charged soap bubble of radius r is in equilibrium with air pressure inside and outside equal. If T is the surface tension of soap solution, then potential of the soap bubble is -

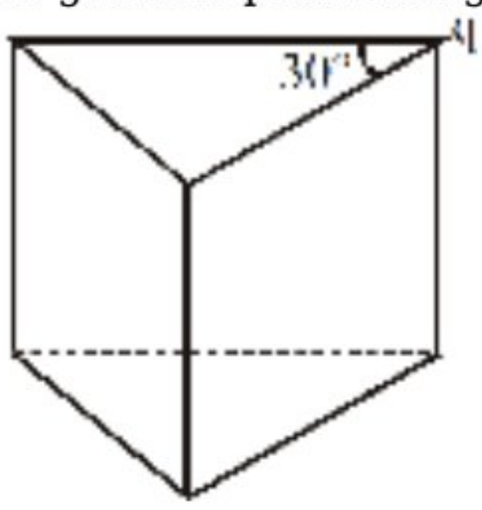
(1) $\sqrt{8T\varepsilon_0/r}$

(2) $\sqrt{8Tr/\varepsilon_0}$

(3) $\sqrt{8Tr\varepsilon_0}$

(4) none of these

20) A prism shaped imaginary structure is given. A point charge is kept as given in figure. Calculate electric flux passing through the prism.



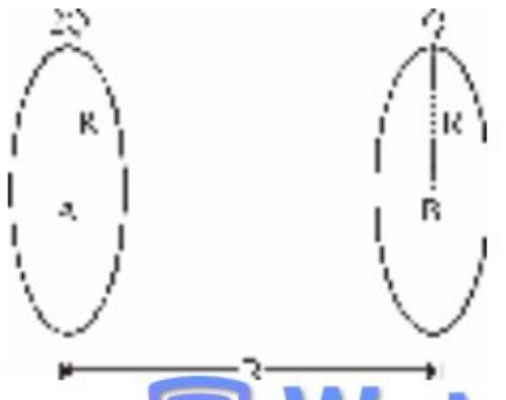
(1) $\frac{q}{4\in_0}$

(2) $\frac{q}{8\in_0}$

(3) $\frac{q}{16\in_0}$

(4) $\frac{q}{24\in_0}$

21) In the following arrangement. Choose correct option:-



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(1) $V_A = \frac{Q(2\sqrt{2}-1)}{4\pi\in_0\sqrt{2}R}$

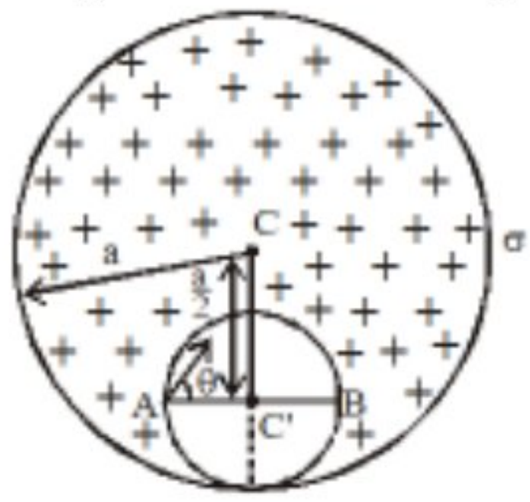
(2) $V_B = \frac{Q(\sqrt{2}-1)}{4\pi\in_0 R}$

W.D. in moving a charge +q from B to A

(3) $W_{BA} = \frac{qQ(3\sqrt{2}-3)}{4\sqrt{2}\pi\in_0 R}$

(4) all options are correct

22) A particle of mass m and charge q is projected with initial velocity v_0 making angle θ with line AB. There is no gravity inside the cavity. Charge is uniformly distributed throughout the volume



except cavity. Path of the particle inside the cavity is :

(1) Straight line

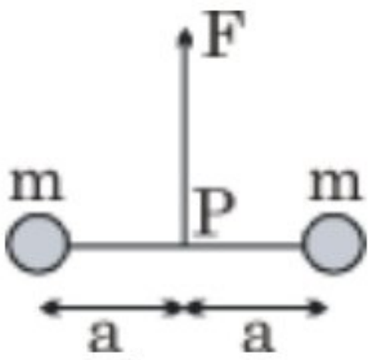
(2) Circle

(3) Parabola

(4) Helical path

23)

Two particles of mass m each are tied at the ends of a light string of length 2a. The whole system is kept on a frictionless horizontal surface with the string held tight so that each mass is at a distance 'a' from the centre P (as shown in the figure). Now, the mid-point of the string is pulled vertically upwards with a small but constant force F. As a result, the particles move towards each other on the surfaces. The magnitude of acceleration, when the separation between them becomes 2x, is



(1) $\frac{F}{2m}\frac{a}{\sqrt{a^2-x^2}}$

(2) $\frac{F}{2m}\frac{x}{\sqrt{a^2-x^2}}$

(3) $\frac{F}{2m}\frac{x}{a}$

(4) $\frac{F}{2m}\frac{\sqrt{a^2-x^2}}{x}$

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24) The maximum electric field intensity on the axis of a uniformly charged ring of charge q and radius R will be :

(1) $\frac{1}{4\pi\varepsilon_0}\frac{q}{3\sqrt{3}R^2}$

(2) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{3R^2}$

(3) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{3\sqrt{3}R^2}$

(4) $\frac{1}{4\pi\varepsilon_0}\frac{3q}{2\sqrt{3}R^2}$

25) A thin spherical conducting shell of radius R has a charge q. A point charge Q is placed at the centre of the shell. The electrostatic potential at a point p a distance R/2 from the centre of the shell is :-

(1) $\frac{(q+Q)}{4\pi\varepsilon_0}\frac{2}{R}$

(2) $\frac{2Q}{4\pi\varepsilon_0 R}$

(3) $\frac{2Q}{4\pi\varepsilon_0 R} - \frac{2q}{4\pi\varepsilon_0 R}$

(4) $\frac{2Q}{4\pi\varepsilon_0 R} + \frac{q}{4\pi\varepsilon_0 R}$

26) Let V_0 be the potential at the origin in a electric field $\vec{E} = (E_x)\hat{i} + (E_y)\hat{j}$. The potential at the point P(x, y) is :

(1) $V_0 - xE_x - yE_y$

(2) $V_0 + xE_x + yE_y$

(3) $xE_x + yE_y - V_0$

(4) $(\sqrt{x^2+y^2})(\sqrt{E_x^2+E_y^2}) - V_0$

27)

Electric field intensity at point 'P' due to the given short charged wire. (Assumed charge on wire = Q)



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(1) $\frac{2K\lambda}{\ell}$

(2) $\frac{K\lambda}{\ell}$

(3) $\sqrt{3}\frac{K\lambda}{\ell}$

(4) $\frac{4K\lambda}{\sqrt{5}\ell}$

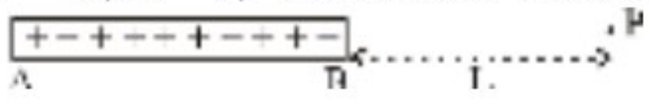
28) Three positive charges of equal value q, are placed at the vertices of an equilateral triangle. The resulting lines of force should be sketched as in:-

(1) 

(2) 

(3) 

(4) 

29) A rod AB of length L having charge density $\lambda = \alpha(2L - x)^2$ where x is distance from end A and α is a constant then electric field at point P will be :- 

(1) K α L

(2) $\frac{K\alpha}{L}$

(3) K α L^2

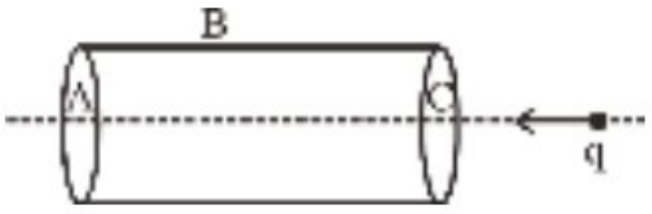
(4) $\frac{K\alpha}{L^2}$

30) Consider a Gaussian surface as shown. A and C are the circular faces while B is the curved surface. A charge q is moved along the axis of cylinder with constant velocity v. Choose the incorrect variation of flux through the surface / cylinder.

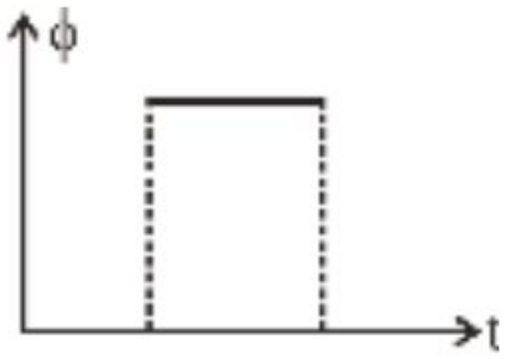
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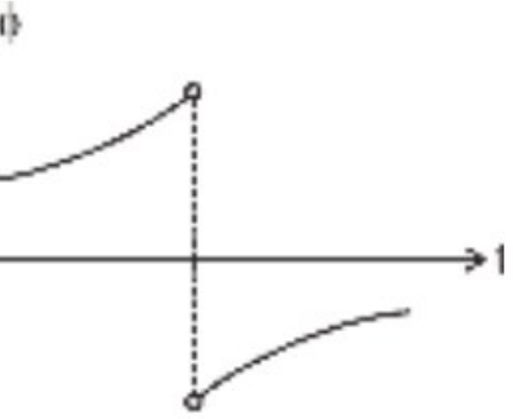
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(1)

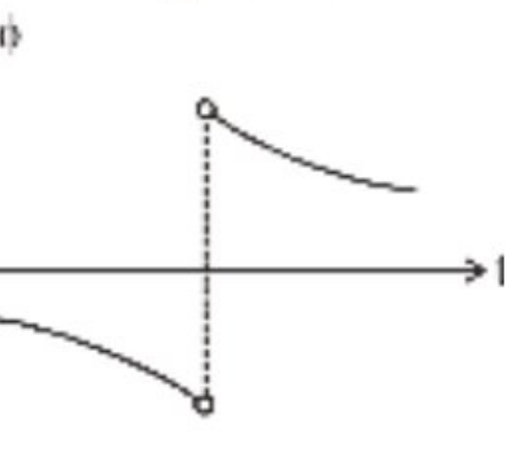


(2)



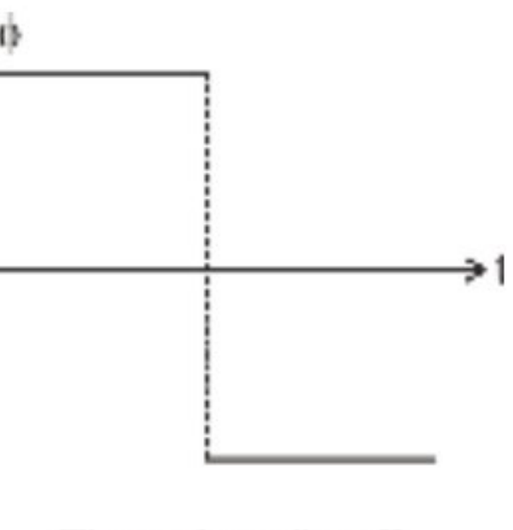
Through surface A

(3)



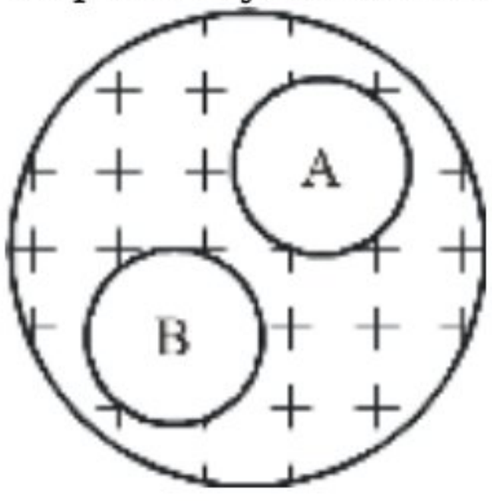
Through surface C

(4)



Through surface B

31) A solid non conducting sphere has two spherical cavities A and B of radius 'r_A' and 'r_B' respectively, as shown. Radius of sphere is 'R'. Select the correct statement :-



(1) Electric field at the centre of cavity A depends on radius of cavity A.

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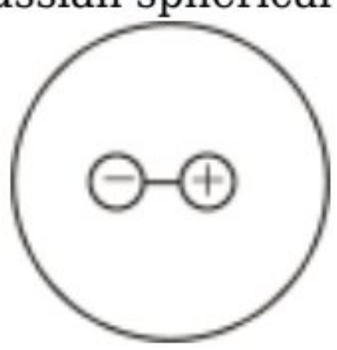
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(2) Electric field at the centre of cavity A will depend on radius of cavity B and distance between centres of cavity A and B.

(3) Electric field in cavity A is uniform.

(4) All of these

32) Consider a gaussian spherical surface, covering a dipole of charge q and –q, then select the



wrong statement.

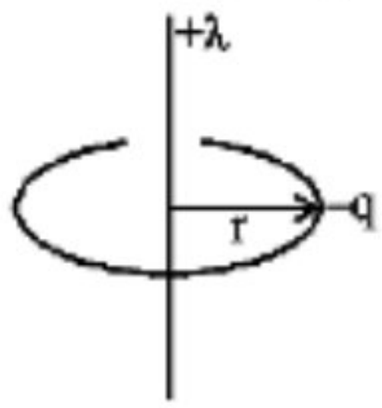
(1) $q_{in} = 0$ (Net charge enclosed by the spherical surface)

(2) $\phi_{net} = 0$ (Net flux coming out the spherical surface)

(3) E = 0 at all points on the spherical surface

(4) $\int \vec{E} \cdot d\vec{s} = 0$ (Surface integral of $\vec{E}$ over the spherical surface)

33) A particle of charge -q and mass m moves in a circle of radius r around an infinitely long line charge of linear charge density +λ. Then time period will be :- $\left(k = \frac{1}{4\pi\varepsilon_0}\right)$.



(1) $T = 2\pi r\sqrt{\frac{m}{2k\lambda q}}$

(2) $T^2 = \frac{4\pi^2 m}{2k\lambda q}$

(3) $T = \frac{1}{2\pi r}\sqrt{\frac{2k\lambda q}{m}}$

(4) $T = \frac{1}{2\pi r}\sqrt{\frac{m}{2k\lambda q}}$

34) Two charges of magnitude $5\,nC$ and $-2\,nC$ are placed at points $(2\,cm,\ 0,\ 0)$ and $(x\,cm,\ 0, 0)$ in the region of space. There is no external field. If electrostatic potential energy of the system is $-0.5\,\mu J$. The value of x is

(1) 10

(2) 20

(3) 30

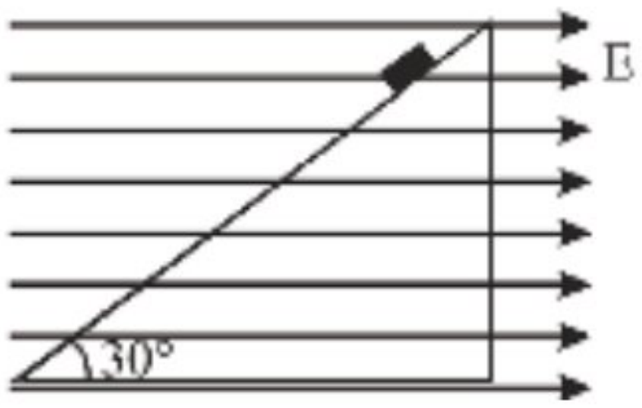
(4) 40

35) A positively charged block of charge 4μC and mass m is placed on inclined plane. A uniform electric field 5×10^4 N/C is applied in the horizontal direction. If the block is in equilibrium, then the

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value of m will be :-

(1) $\frac{20\sqrt{3}\times10^{-2}}{g}$

(2) $\frac{10\sqrt{3}\times10^{-2}}{g}$

(3) $10\sqrt{3}\times10^{-2}$

(4) $20\sqrt{3}\times10^{-2}$

36) A non conducting spherical shell of diameter 10 cm has a charge of 1.6×10^{-4}C. A charge of 20 C is placed at a distance of 10 cm from its centre, then force between them will be approximately :

(1) 4×10^{9} N

(2) 16×10^{13} N

(3) 3×10^{9} N

(4) 6×10^{9} N

37) Charge Q is placed on each of (n –1) corners of a polygon of n sides. The distance of centre of the polygon from each corner is r, then electric field at centre is:

(1) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$

(2) $\frac{(n-1)}{4\pi\varepsilon_0}\frac{Q}{r^2}$

(3) $\frac{n}{(n-1)}\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$

(4) Zero

38) Incorrect statement is - (R = radius of the sphere, r = distance of point of observation from the centre of sphere)

(1) Electric field at a point outside a non conducting solid charged sphere is given by $R^3\rho/3\varepsilon_0r^2$.

(2) Electric field at a point on the surface of non conducting solid charged sphere is given by $R\rho/3\varepsilon_0$.

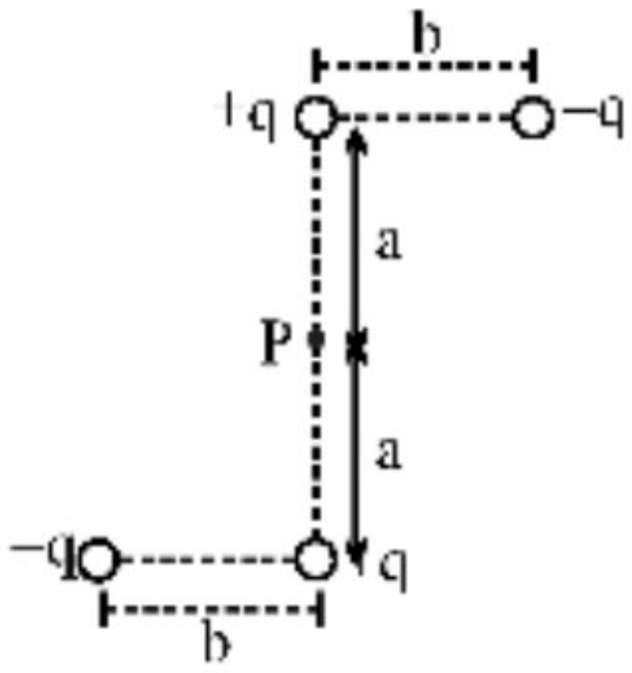
(3) Electric field at a point inside a non conducting solid charged sphere is given by $\rho r/3\varepsilon_0$.

(4) Electric field at a point inside a non conducting solid charged sphere is zero.

39) The electrostatics potential due to the charge configuration at point P as shown in figure for b

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<< a is .

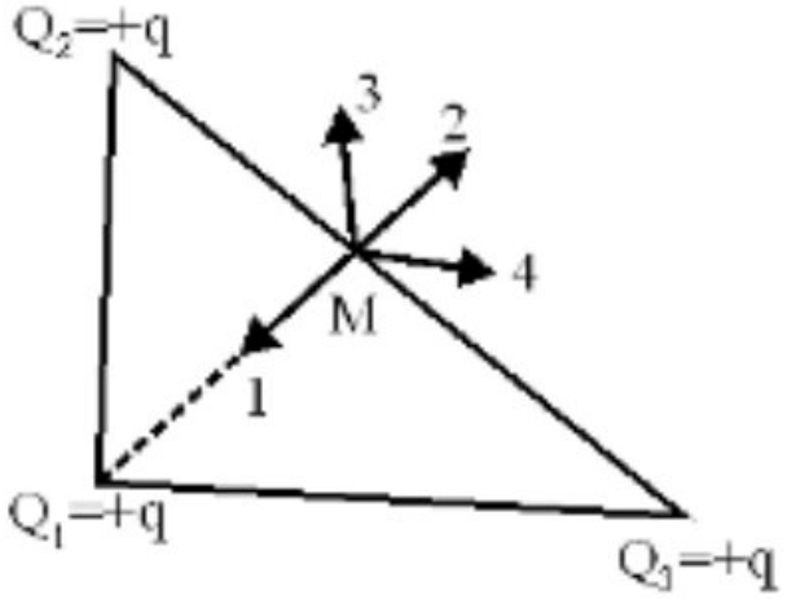
(1) $\frac{2q}{4\pi\varepsilon_0 a}$

(2) $\frac{2qb^2}{4\pi\varepsilon_0 a^3}$

(3) $\frac{qb^2}{4\pi\varepsilon_0 a^3}$

(4) zero

40) Three point charges, as shown, are placed at the vertices of an isosceles right angled triangle. Which of the numbered vectors coincides in direction with the electric field at the mid-point M of the hypotenuse?



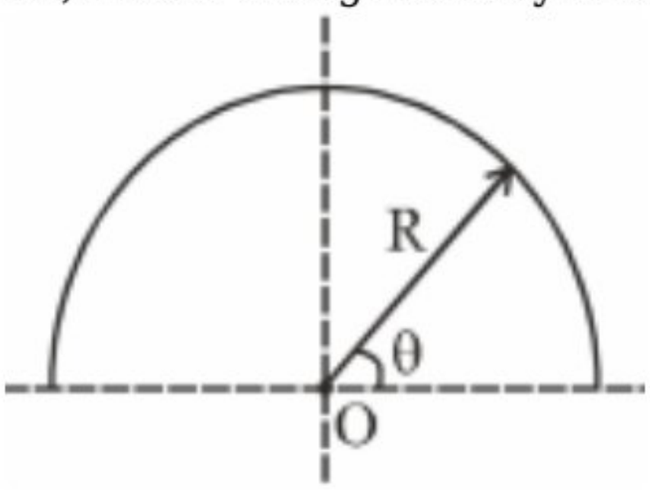
(1) 1

(2) 2

(3) 3

(4) 4

41) Linear charge density of a half ring varies with θ as $\lambda = \lambda_0 \cos\theta$, then find total charge on ring.



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(1) $\lambda_0 R$

(2) $-\lambda_0 R$

(3) $\sqrt{2}\lambda_0 A$

(4) zero

42) A polythene piece rubbed with wool is found to have negative charge of 3.2×10^{-7} C. Calculate the number of electrons transferred from wool to polythene.

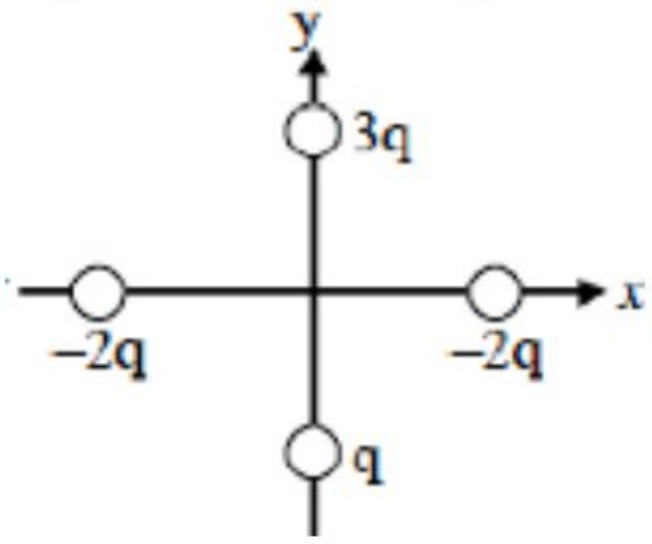
(1) 2×10^{12}

(2) 3×10^{19}

(3) 10^{13}

(4) 10^{18}

43) Four charges are placed each at a distance a from origin. The dipole moment of configuration is



(1) 2q aj

(2) $3q\,a\hat{j}$

(3) $2aq\,[\hat{i}+\hat{j}]$

(4) None

44) A particle of charge –q and mass m moves in a circular orbit of radius r about a fixed charge +Q. The relation between the radius of the orbit r and the time period T is :

(1) $r = \frac{Qq}{16\pi^2 \in_0 m} T^2$

(2) $r^3 = \frac{Qq}{16\pi^3 \in_0 m} T^2$

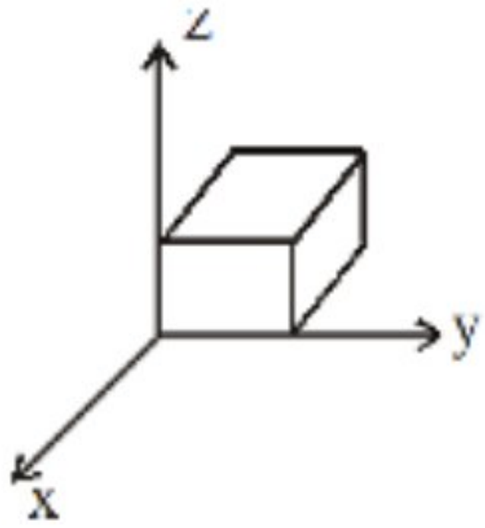
(3) $r^2 = \frac{Qq}{16\pi^3 \in_0 m} T^3$

(4) $r^2 = \frac{Qq}{4\pi^3 \in_0 m} T^3$

45) Figure below shows a closed Gaussian surface in the shape of a cube of edge length 3.0 m. There exists an electric field given by $\vec{E} = [(2.0x + 4.0)i + 8.0j + 3.0k]\,N/C$, where x is in metres, in the region in which it lies. The net charge in coulombs enclosed by the cube is equal to :-

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(1) $-54\in_0$

(2) $6\in_0$

(3) $-6\in_0$

(4) $54\in_0$

CHEMISTRY

1) Normality and molarity of '15V' H_2O_2 solution respectively is :

(1) 1.33 N, 2.66 M

(2) 1.33 M, 1.33 M

(3) 2.66 N, 1.33 M

(4) 3.99 N, 1.33 M

2) An open beaker containing a pure solvent and a second beaker containing a solution in the same solvent with a non-volatile solute, are sealed in a container. Over the period of time :-

(1) The volume of solution increases and the volume of solvent decreases

(2) The volume of both solution and solvent increases

(3) The volume of solution decreases while volume of solvent increases

(4) The volume of both solution and solvent decreases

3) 9.45 gm oxalic acid is present in 250 ml of the solution. How much water should be added to prepare decinormal solution (M. wt. of oxalic acid = 126) :-

(1) 2 litre

(2) 1.5 litre

(3) 1.25 litre

(4) 1 litre

4) Density of a 2.05 M solution of H_2SO_4 in water is 1.08 g/ml. Find its Normality and molality:-

(1) 4.1 N, 2.33 m

(2) 4.1 N, 3.28 m

(3) 4.1 N, 2.88 m

(4) 4.1 N, 4.1 m

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5) The Henry's law constant for the solubility of N_2 gas in water at 298 K is 1.0×10^5 atm. The mole fraction of N_2 in air is 0.8. The number of moles of N_2 from air dissolved in 10 moles of water at 298 K and 5 atm pressure will be

(1) 4.0×10^{-4}

(2) 4.0×10^{-5}

(3) 5.0×10^{-4}

(4) 4.0×10^{-6}

6) According to Henry's law, the partial pressure of gas (P_g) is directly proportional to mole fraction of gas in liquid solution, $P_{gas} = K_H . X_{gas}$, where K_H is Henry's constant. Which is incorrect ?

(1) K_H is characteristic constant for a given gas-solvent system

(2) Higher is the value of K_H, lower is solubility of gas for a given partial pressure of gas

(3) K_H has temperature dependence

(4) K_H decreases with increase of temperature

7) If the total vapour pressure of the liquid mixture A and B is given by the equation:
$P = (180X_A + 90)$ mm Hg
then the ratio of the vapour pressure of the pure state for liquids A and B is given by:

(1) 3 : 2

(2) 4 : 1

(3) 1 : 3

(4) 3 : 1

8) At 40°C the vapour pressure (in torr) of methyl alcohol (A) & ethyl alcohol (B) solution is represented by $P = 120X_A + 138$; where X_A is mole fraction of methyl alchol. The value of $\lim_{x_A \to 0} \frac{P_B^\circ}{X_B}$ & $\lim_{x_B \to 0} \frac{P_A^\circ}{X_A}$ are :-

(1) 138, 258

(2) 258, 138

(3) 120, 138

(4) 138, 125

9) An ideal mixture of liquids A and B with 2 moles of A and 2 moles of B has a total vapour pressure of 1 atm at a certain temperature. Another mixture with 1 mole of A and 3 moles of B has a vapour pressure greater than 1 atm. But if 4 moles of C are added to the second mixture, the vapour pressure comes down to 1 atm. Vapour pressure of C, $P_c^0 = 0.8$ atm. Calculate the vapour pressures of pure A and pure B :-

(1) $P_A^0 = 1.4$ atm, $P_B^0 = 0.7$ atm

(2) $P_A^0 = 1.2$ atm, $P_B^0 = 0.6$ atm

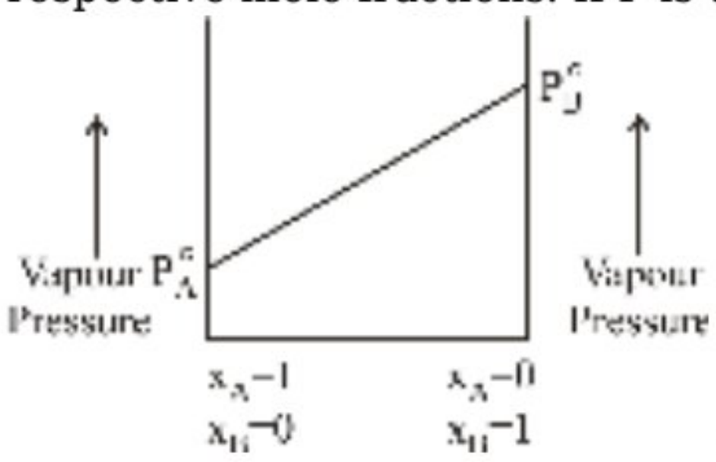
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(3) P_A^0 = 1.4 atm, P_B^0 = 0.6 atm

(4) P_A^0 = 0.6 atm, P_B^0 = 1.4 atm

10) The following is a graph plotted between the vapour pressure of two volatile liquids against their respective mole fractions. If P is the total pressure then which of the following is correct :-



(A) When x_A = 1 and x_B = 0, then P = P_A^0

(B) When x_A = 1 and x_B = 0, then P < P_B^0

(C) When x_B = 1 and x_A = 0, then P = P_B^0

(D) When x_B = 1 and x_A = 0, then P > P_A^0

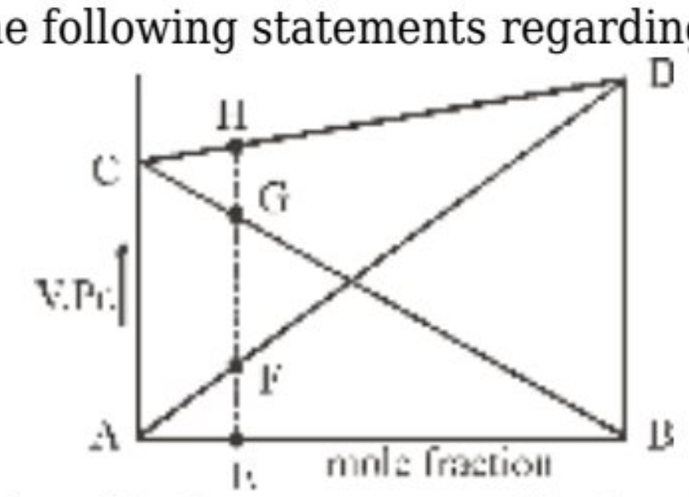
(1) A and B only

(2) C and D only

(3) A, B, C and D

(4) A and C only

11) Based on the given diagram which of the following statements regarding the homogenous solutions of two volatile liquids are correct ?



(1) Plots AD and BC show that Raoult's law is obeyed for the solution in which B is a solvent and A is the solute and as well as for that in which A is solvent and B is solute.

(2) Plot CD show vapour pressure of binary solution of component A and B

(3) EF + EG = EH; and AC and BD correspond to the vapour pressures of the pure solvents A and B respectively.

Select the correct answer using the options given below :-

(1) Only 1

(2) 2 and 3

(3) 1 and 3

(4) All

12) An ideal solution is formed by mixing two volatile liquids A and B. X_A and X_B are the mole fractions of A and B respectively in the solution and Y_A and Y_B are the mole fraction of A and B respectively in the vapour phase. A plot of $\frac{1}{Y_A}$ along y-axis against $\frac{1}{X_A}$ long x-axis gives a straight line. What is the slope of the straight line?

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(1) $\frac{P_B^o}{P_A^o}$

(2) $\frac{P_A^o}{P_B^o}$

(3) $P_B^o - P_A^o$

(4) $P_A^o - P_B^o$

13) Match the column :

Column-I		Column-II	
(A)	n-hexane + n-heptane	(P)	Can be separated by fractional distillation
(B)	Acetone + chloroform	(Q)	Maximum boiling azeotrope
(C)	Chloro-benzene and bromo-benzene	(R)	Cannot be separated by fractional distillation completely
(D)	Ethanol + water	(S)	Minimum boiling azeotrope

Option

(1) A → P, Q ; B → S; C → Q, R ; D → R

(2) A → P ; B → Q, R ; C → P ; D → S, R

(3) A → R ; B → P, R ; C → Q ; D → S, P

(4) A → P, R ; B → Q ; C → Q, R ; D → S

14) The vapour pressure of a solution two liquids, A(P° = 80 mm, X = 0.4) and B(P° = 120 mm, X = 0.6) is found to be 100 mm. it shows that the solution exhibits. (X = mole fraction)

(1) Negative deviation from ideal behaviour

(2) Positive deviation form ideal behavior

(3) Ideal behaviour

(4) Positive deviation at lower concentration

15) How many of following statements are correct.

(a) The unit of ebulioscopic constant is K.kg. mol^{-1} or K $(molality)^{-1}$

(b) Azeotropic mixtures are formed only by non-ideal solutions and they may have boiling points either greater than both the components or less than both the components.

(c) The composition of the vapour phase is same as that of the liquid phase of an azeotropic mixture.

(d)Methanol and acetone mixture will show a positive deviation from Raoult's law

(e)In comparison to a 0.01 M solution of glucose, the depression in freezing point of a 0.01 M $MgCl_2$ solution is about three time(s).

(f)Two different solutions of sucrose of same molality prepared in different solvents will have the different depression in freezing point.

(g)Decreasing order of osmotic pressure for 0.01 M aqueous solutions of barium chloride, potassium chloride, acetic acid and sucrose is $BaCl_2$ > KCl > CH_3COOH > sucrose.

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(h)In osmosis, solvent molecules move through a semipermeable membrane from a region of lower concentration of solute to a region of higher concentration.
(i)The value of molal depression constant depends on nature of solvent.
(j)Relative lowering of vapour pressure, is a dimensionless quantity.

(1) 10
(2) 9
(3) 8
(4) 7

16) The relative lowering of vapour pressure of an aqueous solution containing non volatile solute is 0.0125. The molality of the solution is :-

(1) 0.70
(2) 0.50
(3) 0.60
(4) 0.80

17) pH of a 1 M monobasic acid is found to be 2. Hence its osmotic pressure at a given temperature T K is -

(1) 0.1 RT
(2) 0.11 RT
(3) 1.1 RT
(4) 1.01 RT

18) Assertion (A): Reverse osmosis can occur only when applied pressure is greater than osmotic pressure.
Reason (R): Osmosis naturally occurs from concentrated to dilute solution.

(1) Both A and R are correct, and R explains A
(2) Both A and R are correct, but R does not explain A
(3) A is correct, R is incorrect
(4) A is incorrect, R is correct

19) A solution of urea (mol. mass 60 g mol^{-1}) boils at 100. 18°C at the atmospheric pressure. If K_f and K_b for water are 1.86 and 0.512 K kg mol^{-1} respectively, the above solution will freeze at :-

(1) –6.54°C
(2) 6.54°C
(3) 0.654°C
(4) –0.654°C

20) A 0.001 molal solution of $Pt(NH_3)_4(Cl)_4$ in water had a freezing point depression of 0.0054°C K_f for water is 1.8 the correct formulation for the above molecule is

(1) $[Pt(NH_3)_4Cl_3]Cl$

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(2) $[Pt(NH_3)_4Cl_2]Cl_2$

(3) $[Pt(NH_3)_4Cl]Cl_3$

(4) $[Pt(NH_3)_4Cl_4]$

21)

If $pK_a = -\log K_a = 4$ for a weak acid HX and $K_a = C\alpha^2$ then Van't Haff factor when C = 0.01 M is :–

(1) 1.01

(2) 1.02

(3) 1.10

(4) 1.20

22) Relative decrease in vapour pressure of an aqueous NaCl solution is 0.167, number of moles of NaCl present in 180gm of H_2O is :-

(1) 2 mol

(2) 1 mol

(3) 3 mol

(4) 4 mol

23) If relative decrease in vapour pressure is 0.064 for a solution, containing, 0.1 mol NaCl in 3 mol H_2O, NaCl is ..... ionized (approx) :

(1) 60%

(2) 50%

(3) 100%

(4) 40%

24) 0.5 mole Na_2SO_4 is dissolved in 500 gm H_2O, then determine freezing point and boiling point of the solution- (Given:- $(k_f)_{H_2O} = 1.8K - kg/mol (k_b)_{H_2O} = 0.5K - kg/mol$)

(1) –1.8°C, 100.5°C

(2) –5.4°C, 101.5°C

(3) –5.4°C, 98.5°C

(4) 0°C, 100°C

25) A solution of a substance containing 1.05 g per 100 mL. was found to be isotonic with 3%(w/v) glucose solution. The molecular mass of the substance is :

(1) 31.5

(2) 6.3

(3) 630

(4) 63

26) For the reaction R → P, the concentration of a reactant changes from 0.03 M to 0.02 M in 25

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minutes. Average rate of reaction in Ms^{-1} is :-

(1) 0.4×10^{-3} mol $L^{-1}s^{-1}$

(2) 6.6×10^{-6} mol $L^{-1}s^{-1}$

(3) 0.66×10^{-4} mol $L^{-1}s^{-1}$

(4) 4×10^{-5} mol $L^{-1}s^{-1}$

27) Rate law for the reaction, $A + B \rightarrow$product is rate = $k[A]^2[B]$. What is the rate constant, if rate of reaction at a given temperature is 0.22 Ms^{-1}, when [A] = 1M and [B] = 0.25 M ?

(1) 3.52 $M^{-2}s^{-1}$

(2) 0.88 $M^{-2}s^{-1}$

(3) 1.136 $M^{-2}s^{-1}$

(4) 0.05 $M^{-2}s^{-1}$

28)

The rate of a reaction is expressed as:

$$+\frac{1}{2}\frac{\Delta[C]}{\Delta t} = \frac{1}{3}\frac{\Delta[D]}{\Delta t} = \frac{1}{4}\left(-\frac{\Delta[A]}{\Delta t}\right) = \left(-\frac{\Delta[B]}{\Delta t}\right)$$

The reaction is

(1) $4A + B \rightarrow 2C+3D$

(2) $B + 3D \rightarrow 4A+2C$

(3) $A + B \rightarrow C+D$

(4) $B + D \rightarrow A+C$

29) The initial rate of reaction $A + 5B + 6C \rightarrow 3L + 3M$ has been determined under the following conditions :-

Experiment No.	**$[A]_0$ (in M)**	**$[B]_0$ (in M)**	**$[C]_0$ (in M)**	**Initial rate (in M min^{-1})**
1.	0.02	0.02	0.02	2.08×10^{-3}
2.	0.01	0.02	0.02	1.04×10^{-3}
3.	0.02	0.04	0.02	4.16×10^{-3}
4.	0.02	0.02	0.04	8.32×10^{-3}

Determine the rate law ?

(1) Rate = $k[A]\,[B]\,[C]^2$

(2) Rate = $k[A]\,[C]^2$

(3) Rate = $k[A]\,[B]\,[C]$

(4) Rate = $k[A]\,[B]^5[C]^6$

30) The hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below –

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$A_2 \rightleftharpoons A + A$ ..... (fast)
$A + B_2 \rightarrow AB + B$ ..... (slow)
$A + B \rightarrow AB$ ..... (fast)
The order of the over all reaction is –

(1) 2

(2) 1

(3) 1½

(4) Zero

31) What is order of gaseous reaction with respect to $H_{2(g)}$

$$2NO_{(g)} + 2H_{2(g)} \rightarrow N_{2(g)} + 2H_2O_{(g)}$$

If this reaction occurs in following steps

$$2NO \underset{K_B}{\overset{K_F}{\rightleftharpoons}} N_2O_2$$

(i) (fast step)

(ii) $N_2O_2 + H_2 \rightarrow N_2O + H_2O$ (slow step)

(iii) $N_2O + H_2 \rightarrow N_2 + H_2O$ (fast step)

(1) 3

(2) 2

(3) 1

(4) 0

32) For a reversible reaction, $2NH_3 \underset{K_2}{\overset{K_1}{\rightleftharpoons}} N_2 + 3H_2$;
what will be rate of Appearance of H_2 ?

(1) $K_1[NH_3]^2 - K_2[N_2][H_2]^3$

(2) $3K_1[NH_3]^2 - 3K_2[N_2][H_2]^3$

(3) $3K_1[2NH_3]^2 - 3K_2[N_2][3H_2]^3$

(4) $K_1[NH_3] - K_2[N_2][H_2]$

33) The activation energies of two reactions are E_1 and E_2 with $E_1 > E_2$. If the temperature of the reacting system is increased from T_1(rate constants are K_1 and K_2) to T_2(rate constants are K_1' and K_2') predict which of the following alternative is correct ?

(1) $\frac{K_1'}{K_1} = \frac{K_2'}{K_2}$

(2) $\frac{K_2'}{K_1} > \frac{K_2'}{K_2}$

(3) $\frac{K_1'}{K_1} < \frac{K_2'}{K_2}$

(4) $K_1' < K_2'$

34)

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NOVA

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A reaction take place in various steps. The rate constant of overall reaction is given as

$$K = \frac{K_1}{K_3}\left(\frac{K_2}{K_4}\right)^{1/2}$$

If the activation energies are 60, 40, 50 and 10 KJ/mol for different steps respectively. Then overall activation energy will be :

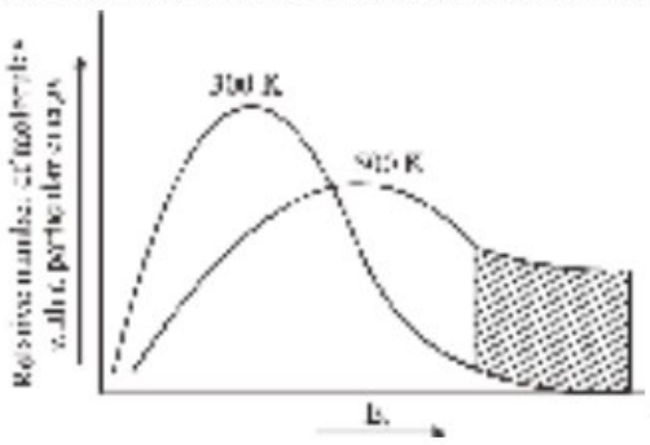
(1) 10 KJ

(2) 20 KJ

(3) 30 KJ

(4) 25 KJ

35) The distribution of molecular kinetic energy at two temperatures is as shown in the following graph :



Which of the following conclusions is/are correct?

(A) The number of molecules with energy E_a or greater is proportional to the shaded area for each temperature

(B) The number of molecules with energy E_a or less is proportional to the shaded area of each temperature

(C) The number of molecules with energy E_a is the mean of all temperatures

(D) The graph follows the Maxwell-Boltzmann energy distribution law

(1) only A

(2) only B

(3) only C

(4) A and D

36) A hydrogenation reaction is carried out at 500K. If the same reaction is carried out in the presence of a catalyst at the same rate with same frequency factor, the temperature required is 400 K. What is the activation energy of the reaction, at 500 K if the catalyst lowers the activation energy barrier by 16 kJ/mole.

(1) 100 kJ/mole

(2) 80 kJ/mole

(3) 60 kJ/mole

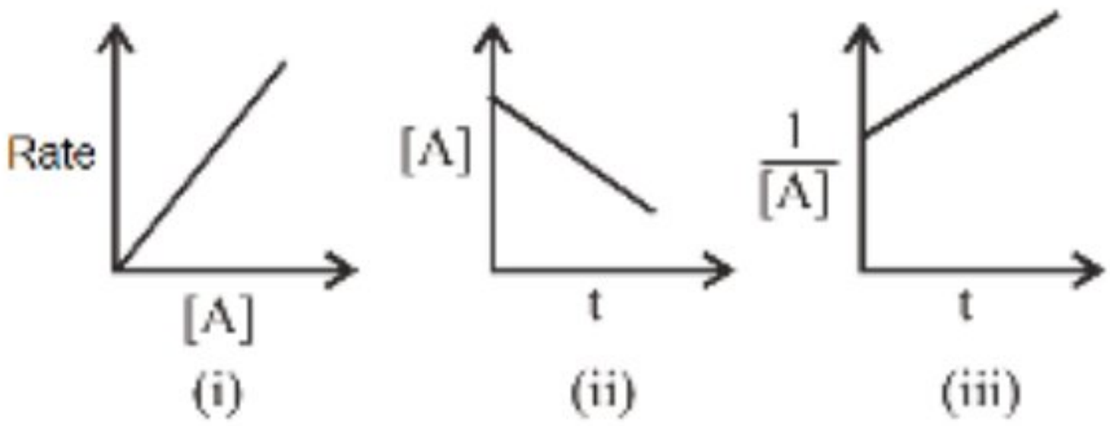
(4) None of these

37) Consider the plots for the types of reaction .

$$nA \rightarrow B + C$$

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These plots respectively corresponds to the reaction orders :

(1) 0, 2, 1

(2) 0, 1, 2

(3) 1, 1, 2

(4) 1, 0, 2

38) For a reaction A —→ Product, half-life measured for two different values of initial concentrations 5×10^{-3} M and 25×10^{-4} M are 1.0 and 2.0 hrs respectively. If initial concentration is adjusted to 1.25×10^{-3} M, the new half-life would be :-

(1) 16 hrs

(2) 32 hrs

(3) 64 hrs

(4) 4 hrs

39) The value of $\left[\frac{t_{0.875}}{t_{0.50}}\right]$ for n^{th} order reaction A → P is :– [Given : $n \neq 1$]
[t_α represents time required for fraction 'α' of reactant to react]

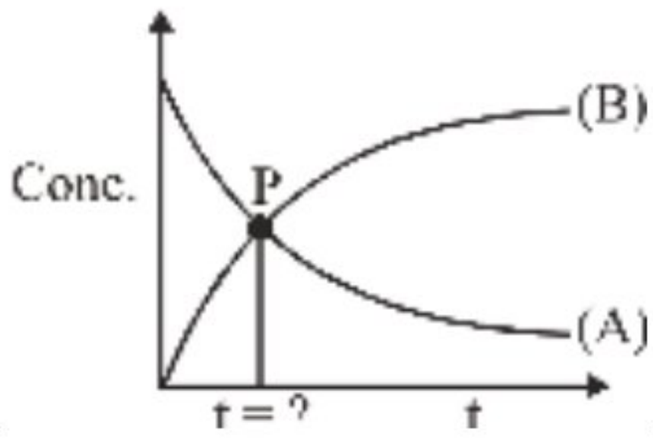
(1) 2^{2n-2}

(2) 2^{2n-1}

(3) $\frac{8^{n-1}-1}{2^{n-1}-1}$

(4) $\frac{4^{n-1}-1}{2^{n-1}-1}$

40) For the reaction 2A → B following graph is obtained



At point 'P', t is equal to :-

(1) $t_{1/3}$

(2) $t_{2/3}$

(3) $t_{1/2}$

(4) None

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41) A second order reaction shows 40% completion in 30 minutes then its half life will be :-

(1) 20 min.
(2) 40 min.
(3) 45 min.
(4) 60 min.

42) Two substances A and B are present such that $[A]_0 = 4[B]_0$ and half-life of A is 5 min and that of B is 15 min. If they start decaying at the same time following first order kinetics, how much time later will the concentration of both of them would be same?

(1) 15 min
(2) 10 min
(3) 5 min
(4) 12 min

43) $SO_2Cl_2 \rightarrow SO_2 + Cl_2$ is a first order gaseous reaction with $K = 2.3 \times 10^{-5}$ sec^{-1} at 320°C. The % of SO_2Cl_2 decomposed on heating for 90 minute is : ($10^{0.054} = 1.13$)

(1) 11.5
(2) 1.15
(3) 0.118
(4) 18.11

44) What is true about pseudo first order reaction?

(1) One of the two reactants is in excess in the reaction
(2) There is only one reactant in the reaction
(3) It is actually a zero order reaction
(4) None of the above

45)

A reaction involving two different reactant can never be

(1) 1st order reaction
(2) Unimolecular reaction
(3) Bimolecular reaction
(4) 2nd order reaction

BIOLOGY

1) Perispermic seeds are formed in :-

(1) Black pepper
(2) Sugar beet

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(3) Litchi

(4) Both (1) and (2)

2) If the haploid number in a flowering plants is 14. What will be the number of chromosomes in integuments, antipodal cells, embryo, endosperm and nucellus respectively -

(1) 14, 28, 7, 42, 21

(2) 7, 14, 42, 28, 14

(3) 28, 14, 28, 42, 28

(4) 28, 42, 14, 28, 14

3) Which of the following statements is wrong -

(1) In commelina, seed-set is assured even in the absence of pollinators

(2) Transfer of pollen grains from anther of a plant to stigma of other plant of same species is xenogamy

(3) Pollen-pistil interaction is mediated by chemicals

(4) Pollen-pistil recognition is always followed by inhibition of pollen germination

4) Read the following statements -
A. The pollen grains are spherical, measuring about 20-25 mm in diameter.
B. The veriety of pollen grain architecture includes colours and designs but not size and shape.
C. Pollen grains are well preserved as fossils because of the presence of chitin.
Select the option which is true for them -

(1) All A, B, C are correct

(2) All A, B, C are incorrect

(3) Only A and B are correct

(4) Only B and C are correct

5)

Select the erroneous (wrong) set of items/terms with respect to ploidy level ?

(1) Nucellus & integuments

(2) Egg & perisperm

(3) Perisperm & pollen mother cell

(4) Pericarp & perisperm

6) Select erroneous pair ?

(1) Perispermic seed - black pepper

(2) Winged pollen grain - *Pinus*

(3) Seed with perisperm, endosperm and caruncle - castor

(4) Five free stigmas - poppy

7) Choose the correct statement from the following:

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(1) Cleistogamous flowers always exhibit autogamy

(2) Chasmogamous flowers always exhibit geitonogamy

(3) Cleistogamous flowers exhibit both autogamy and geitonogamy

(4) Chasmogamous flowers never exhibit autogamy

8) (a) Synergids → Diploid cells
(b) Filiforus apparatus → Female gametophyte
(c) Autogamy → No synchrony needed between poller release & stigma receptivety
(d) Egg apparatus → One egg + 2 polar nuclei
(e) Monosporic development → Female gametophyte developed from two functional megaspores
(f) Nucellus →Abundant food reserve located in female gametophyte
Options :

(1) a, c, f are correct

(2) b, d, e are correct

(3) Only a & f are correct

(4) a, c, d, e and f are wrong

9) Find out the total number of male gametes developed from 70 micro spore mother cell.

(1) 120

(2) 280

(3) 460

(4) 560

10) Which of the following is incorrect for wind-pollinated plants?

(1) Well exposed stamens and stigma

(2) Many ovules in each ovary

(3) Flowers are small and not brightly coloured

(4) Pollen grains are light and non-sticky

11) Large number of seeds are present in fruits of :-

(1) Wheat

(2) *Orobanche*

(3) Maize

(4) Banana

12) Which of the following statements correctly describes the cells found in a mature pollen grain?

(1) The vegetative cell is small with a dense cytoplasm and a nucleus, while the generative cell is larger with abundant food reserve.

(2) Both the vegetative and generative cells are of the same size and shape.

(3) The generative cell is spindle-shaped and floats in the cytoplasm of the vegetative cell, which is larger with abundant food reserve.

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(4) The vegetative cell is spindle-shaped and floats in the cytoplasm of the generative cell, which is larger with abundant food reserve.

13) Given below are two statements: Statement I: In angiosperms, the seed is often described as a fertilised ovule and is formed outside the fruits.
Statement II: The cotyledons of the embryo are simple structures, generally thick and swollen due to storage of food reserves. In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false.
(2) Both Statement I and Statement II are true.
(3) Both Statement I and Statement II are false.
(4) Statement I is false but Statement II is true.

14) **Assertion (A):** In Vallisneria, the short stalk female flowers reach the surface of the water to facilitate pollination.
Reason (R): The male flowers or pollen grains of Vallisneria are released inside the surface of the water and are carried to the female flowers by water currents.

(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) Both A and R are false

15) In Angiosperm, formation of endosperm always precedes development of the ..... :-

(1) Embryo
(2) Embryosac
(3) Male gametophyte
(4) Female gametophyte

16) Dicliny is the contravince of :-

(1) Capsella
(2) Chinarose
(3) Catharanthus
(4) Cucurbita

17) In monocots, mature male gametophyte is :-

(1) 7 celled structure
(2) 3 celled structure
(3) 8 nucleated structure
(4) 2 nucleated structure

18) All of the following include out breeding devices except:

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(1) Bisexuality (Monocliny), Homogamy, Cleistogamy.
(2) Anthers and stigma are placed at different position.
(3) Pollen release and stigma receptivity are not synchronised.
(4) Unisexuality of flowers, self incompatibility.

19) Pollination with the help of water takes place in:-

(1) Triticum and Hordeum
(2) Pisum and Helianthus
(3) Helianthus and Vallisneria
(4) Vallisneria and Hydrilla

20) Which of following layer helps in dehiscence of anther for liberation of pollen grains?

(1) Epidermis
(2) Endothecium
(3) Middle layer
(4) Endothelium

21) Which of the following statement is **wrong**.

(1) Pollen grain in some plants remains viable for months.
(2) Tapetum nourishes the developing pollen
(3) Vegetative cell is larger than generative cell.
(4) PEN is produced by fusion of two polar nuclei

22) Some angiosperm produces more than one embryo in their seed. This phenomenon is called.....discovered by .....

(1) Apomixis, Nawaschin
(2) Polyembryony, Anton Von Leeuwenhoek
(3) Diplospory, Anton Von Leeuwenhoek
(4) Aril, Amici

23) Chalaza end represent -

(1) Basal part of ovule
(2) Apical part of ovule
(3) Basal part of ovary
(4) Apical part of ovary

24) In angiosperms, the haploid, diploid and triploid structures are respectively

(1) Pollen, embryo and endosperm
(2) Male gamete, endosperm and zygote
(3) Scutellum, embryo and aleurone layer

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(4) Aleurone layer, coleoptile and coleorhiza

25) Find the odd one with respect to hydrophilous plants :-

(1) *Vallisneria*
(2) *Hydrilla*
(3) *Zostera*
(4) Grasses

26) Which of these will ensure autogamy and self-pollination to occur in an angiosperm?

(1) Dioecious nature with the plants present in the same area
(2) Monoecious nature with the male and female flowers present close to one another
(3) Cleistogamous nature of bisexual flowers with synchronized maturity of androecium and gynoecium
(4) All of 1, 2 and 3

27) Consider the following statements and choose the correct option.
(a) Proximal end of filament is attached to the thalamus or petal.
(b) Tapetum helps in dehiscence of anther.
(c) In wheat, pollen grains maintain viability for months.
(d) Two anther lobes are attached together with the help of a region, called connective.

(1) (a) and (b)
(2) (a) and (d)
(3) (c) and (d)
(4) (b) and (c)

28) In general the embryo sac in angiosperms consists of :

(1) One egg cell, two synergid cells, two antipodal cells, two polar nuclei.
(2) One egg cell, two synergid cells, two antipodal cells, three polar nuclei.
(3) One egg cell, two synergid cells, three antipodal cells, two polar nuclei
(4) One egg cell, three synergid cells, two antipodal cells, two polar nuclei

29) **Statement - (I) :-** Pollen grains of some plant cause severe allergies and broncial afflictions in some people.
Statement - (II) :- Pollen grains are rich in nutrients and they are used in the form of tablets and syrups.

(1) Both Statement - I & II are incorrect.
(2) Both Statement - I & II are correct.
(3) Statement - I is correct & II is incorrect.
(4) Statement - I incorrect & II is correct.

30) **Assertion (A):** Most zygotes divide only after certain amount of endosperm is formed.

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Reason (R): This is an adaptation to provide assured nutrition to the developing embryo.

(1) Both (A) and (R) are correct but (R) is the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

(4) (A) is not correct but (R) is correct

31) Seeds retain a part of endosperm as it is not completely used up during embryo development, is called -

(1) Non-albuminous seeds

(2) Albuminous seeds

(3) Non-endospermic seeds

(4) Perispermic seeds

32) In which part of the embryo maximum growth take place in epigeal germination.

(1) Radicle

(2) Plumule

(3) Epicotyl

(4) Hypocotyl

33) Number of meiotic divisions required to produce 200 seeds of pea would be:

(1) 200

(2) 400

(3) 300

(4) 250

34) Which of the following is best example of insitu germination ?

(1) Hypogeal

(2) Epigeal

(3) Vivipary

(4) Ex-situ

35) Fill in the blank space with the correct option :-

Pollination by _____ is rare is flowering plants.

(1) Wind

(2) Insect

(3) Water

(4) Birds

36) In bisexual flower when the gynoecium mature earlier than the androecium it is called

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(1) Protandry
(2) Protogyny
(3) Heterogamy
(4) Autogamy

37) After culturing the anther of a plant a few diploid plants were found along with haploid plants. The diploid plant could have arisen from:

(1) Generative cell of pollen
(2) Cell of anther wall
(3) Vegetative cell of pollen
(4) Exine of pollen wall

38) Coleoptile and coleorhiza are found in :-

(1) Castor
(2) Pea
(3) Bean
(4) Grass family

39) In majority of flowering plants functional megaspore and megaspore mother cell are present at the :-

(1) Micropylar end and chalazal end respectively
(2) Chalazal end and micropylar end respectively
(3) Chalazal end
(4) Micropylar end

40) Monoecious plants are :-

(1) If both male and female flowers are present in same plant but flowers are bisexual
(2) If both male and female flowers are present in same plant but flower are unisexual
(3) If both gynoecium and Androecium are present on same flower
(4) If male and female flower are present on different plant but flower are unisexual

41) Sequence of development during the formation of embryosac is :-

(1) Archesporial cell → megaspore → megaspore mother cell → embryosac
(2) Megasporocyte → archesporial cell → megaspore → embryosac
(3) Megaspore → megaspore mother cell → Archesporial cell → embryosac
(4) Archesporial cell → megaspore mother cell → megaspore → embryosac

42) Caruncle is :-

(1) Out growth of funicle
(2) Out growth of integument near chalaza

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(3) Unereness on the surface of endosperm
(4) Out growth of integument near micropyle

43) Flowers are bisexual in :-

(1) Maize
(2) Papaya
(3) Chinarose
(4) Date palm

44) Multicarpellary apocarpous condition of gynoecium is present in :-

(1) Rose
(2) Lotus
(3) *Michelia*
(4) All of the above

45) Which of the following character is common in both monocotyledonous and dicotyledonous embryo ?

(1) Presence of coleorrhiza
(2) Presence of coleoptile
(3) Presence of hypocotyl
(4) Presence of epiblast

46) Read the given statements and choose correct option about them :-
(A) Each lobule of testis contains one to three highly coiled seminiferous tubules and it is lined inside by only columnar cells
(B) The vasa efferentia leave the testis and open into vas deferens
(C) Functional maturation of sperms take place in epididymis.
(D) Seminal plasma is rich in fructose, calcium and certain enzymes.

	A	B	C	D
(1)	True	False	True	True
(2)	False	False	True	True
(3)	False	False	False	False
(4)	True	True	False	False

(1) 1
(2) 2
(3) 3
(4) 4

47) Identify the correct pair :-

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Hormones		Function	
I	Inhibin	a	Development of secondary sexual characters of male
II	Androgen binding protein (ABP)	b	Stimulates Leydig cells to secrete testosterone
III	Testosterone	c	Concentrates testosterone inside seminiferous tubule
IV	LH	d	Suppression of synthesis and release of FSH

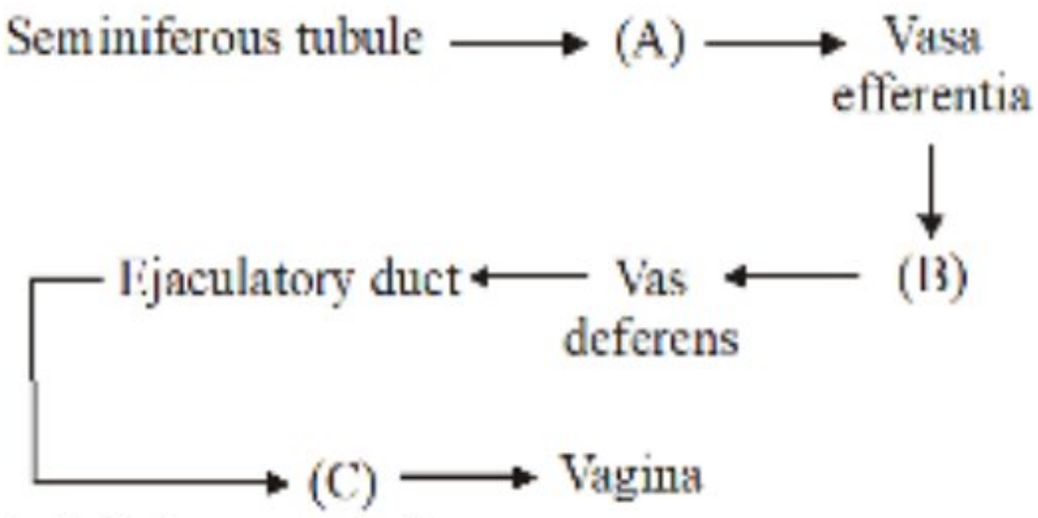
(1) I-d, II-b, III-a, IV-c

(2) I-d, II-c, III-a, IV-b

(3) I-c, II-d, III-a, IV-b

(4) I-c, II-b, III-a, IV-d

48) Here path of sperms in mammals is shown :-



A, B & C respectively are -

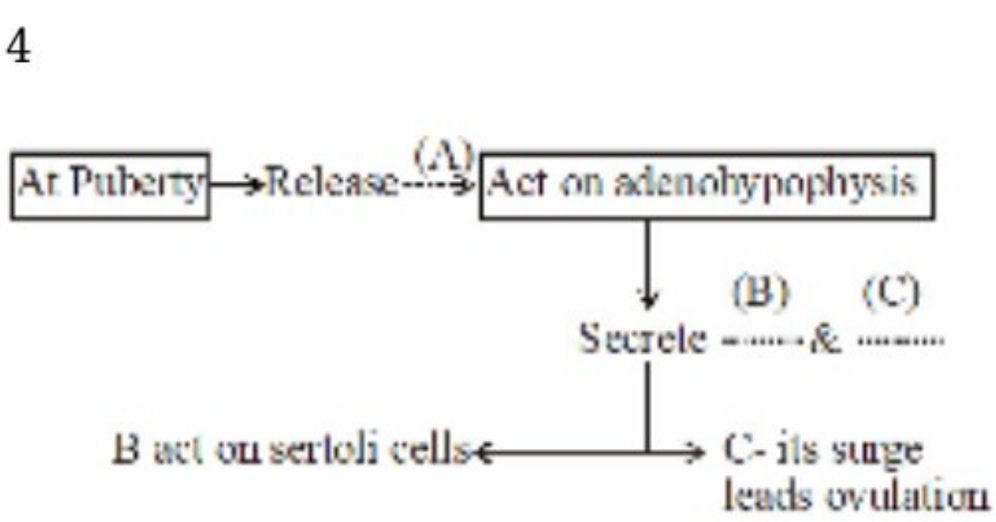
	A	B	C
(1)	Epididymis	Testes	Urethra
(2)	Rete testis	Epididymis	Urethra
(3)	Ovary	Epididymis	Uterus
(4)	Testes	Rete testis	Epididymis

(1) 1

(2) 2

(3) 3

(4) 4



49)

identify the A, B and C :-

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	A	B	C
(1)	GnRH	LH	FSH
(2)	GnRH	FSH	LH
(3)	Somatotrophic	ICSH	STH
(4)	Somatotrophic	Androgen	STH

(1) 1

(2) 2

(3) 3

(4) 4

50) Erection of penis occurs due to :-

(1) Sympathetic nervous system

(2) Parasympathetic nervous system

(3) Both (1) and (2)

(4) Cerebral cortex ® ANS ® Vasoconstriction

51) Why is the presence or absence of hymen not a reliable indicator of virginity or sexual experience?

(1) In some women, the hymen persists even after performing coitus

(2) Hymen can also be ruptured by sudden fall

(3) Hymen may be either present or absent in female

(4) All of the above

52) Which of the following structures are not included in external genitalia of female ?

(a) Mons pubis
(b) Labia majora
(c) Hymen
(d) Labia minora
(e) Cervix
(f) Fallopian tube

(1) c, e and f

(2) Only e and f

(3) Only d and f

(4) a, d, e and f

53) Find out the correct match from the following table.

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	Column I	Column II	Column III
(i)	Primary Spermatocyte	(n)	Maturation phase
(ii)	Secondary Spermatocyte	(n)	Maturation phase
(iii)	Spermatid	(n)	Multiplication phase

(1) (i) only

(2) (i) and (ii)

(3) (ii) only

(4) (ii) and (iii)

54) Read the following statements carefully.

The .....(A)..... present on the inside wall of .....(B)..... multiply by .....(C)..... division. A .....(D)..... completes the .....(E)..... division leading to formation of two equal, haploid cells called .....(F)..... .

Which one of the following options gives the correct filling of A, B, C, D, E and F ?

	A	B	C	D	E	F
(1)	Spermato-gonia	Semini-ferous tubule	Mitotic	Primary Spermato-cyte	First meiotic	Secondary Spermato-cyte
(2)	Primary Spermato-cyte	Sertoli Cell	Mitotic	Secondary Spermato-cyte	First meiotic	Spermatid
(3)	Spermato-gonia	Sertoli Cell	First meiotic	Secondary Spermato-cyte	Second meiotic	Spermatid
(4)	Primary Spermato-cyte	Semini-ferous tubule	First meiotic	Spermatid	Second meiotic	Sperm

(1) 1

(2) 2

(3) 3

(4) 4

55) How many structures in the list given below are haploid ?

Spermatid, secondary oocyte, primary spermatocyte, ovum, sperm, oogonia, spermatogonia, polar body

(1) Six

(2) Four

(3) Two

(4) Five

56) If germ cells in female and male gonads begin undergoing meiosis simultaneously. What will be

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ratio of ova and sperms produced?

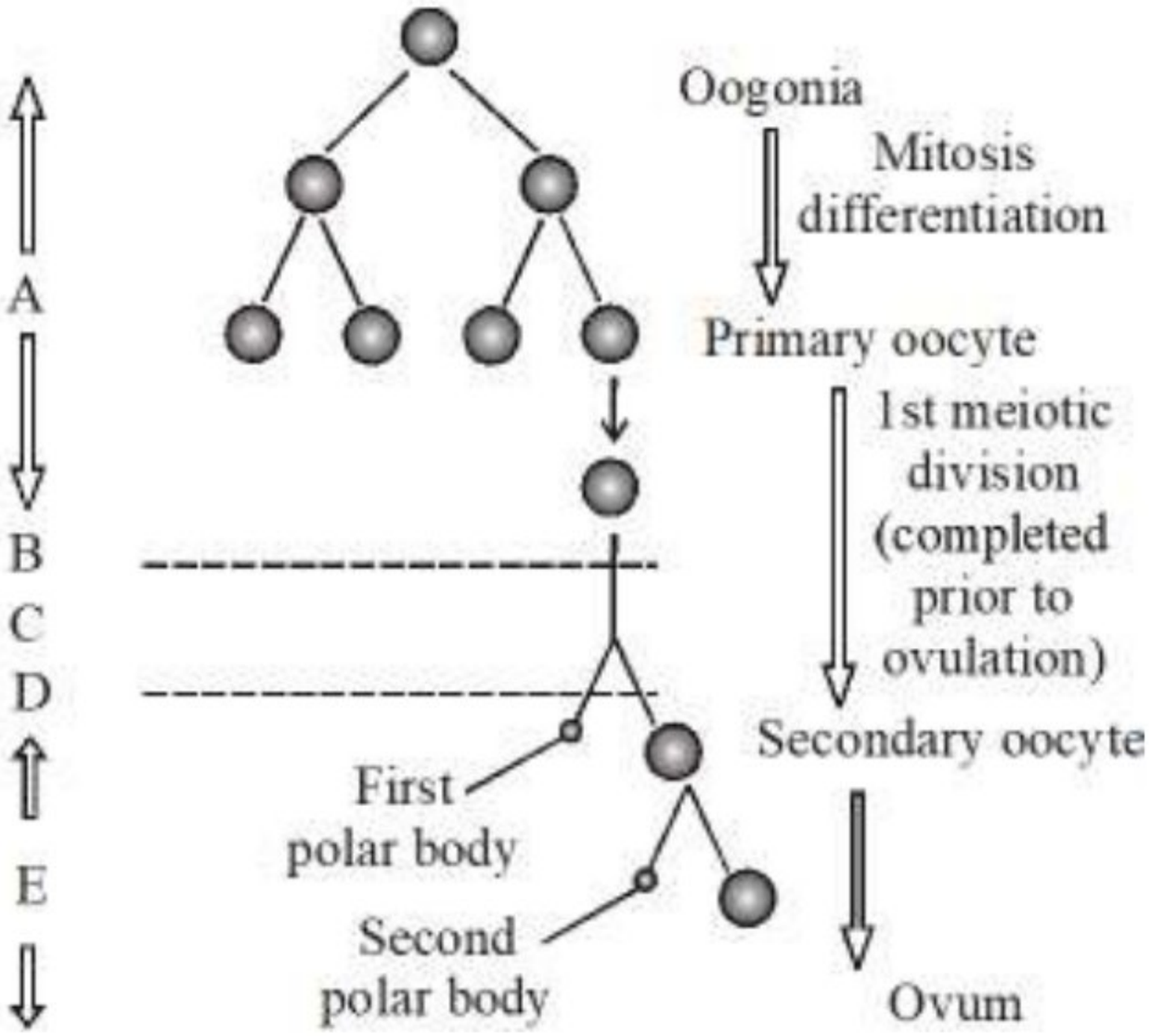
(1) 1 : 1

(2) 1 : 2

(3) 1 : 4

(4) 2 : 1

57) The following refers to schematic representation of oogenesis.
Identify A to E correctly :



(1) A - Fetal life, B - Birth, C - Puberty, D - Adult reproductive life, E - Child body

(2) A - Fetal life, B - Birth, C - Child hood, D - Puberty, E - Adult reproductive life

(3) A - Adult reproductive life, B - Birth, C - Puberty, D - Child hood, E - Fetal life

(4) A - Birth, B - Child hood, C - Fetal life, D - Puberty, E - Adult reproductive life

58) Match list-I (Cell type) with list-II (Characteristic) and select the option with correct matchings.

List-I		List-II	
A.	Spermatogonium	(i)	Does not divide
B.	Spermatocyte	(ii)	Divides meiotically
C.	Spermatid	(iii)	Divides mitotically
D.	Sertoli cell	(iv)	Divides both mitotically and meiotically
		(v)	Nourishes other cell type

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	A	B	C	D
(1)	(v)	(iii)	(i)	(iv)
(2)	(i)	(iii)	(iv)	(v)
(3)	(iii)	(ii)	(iv)	(i)
(4)	(iii)	(ii)	(i)	(v)

(1) 1

(2) 2

(3) 3

(4) 4

59) How many of the following are diploid structures ?
Primary oocyte, spermatozoa, spermatid, oogonium, ovum, second polar body, secondary spermatocyte, spermatogonium.

(1) 4

(2) 5

(3) 3

(4) 6

60) Level of which hormone reaches peak during follicular phase of menstrual cycle ?

(1) Estrogen

(2) Progesterone and estrogen

(3) Luteinizing hormone and hCG

(4) F.S.H. and progesterone

61) How many statement are correct :-
(A) Corpus luteum secretes large amount of progesterone which is essential for the maintenance of endometrium.
(B) Maintenance of endometrium is necessary for implantation of fertilised ovum and other events of pregnancy.
(C) During pregnancy all events of menstrual cycle stop.
(D) In absense of fertilization, the corpus luteum degenerates this cause disintegration of the endometrium leading to menstruation.

(1) 1

(2) 2

(3) 3

(4) 4

62) Arrange following events of female reproductive cycle in (human) in sequence :-
(I) Secretion of FSH
(II) Growth of corpus luteum
(III) Growth of follicles
(IV) Ovulation

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(V) Sudden increase in level of LH.
(VI) Formation of corpus albicans

(1) III → I → IV → II → VI → V
(2) I → III → V → IV → II → VI
(3) I → IV → III → V → II → VI
(4) II → I → III → IV → VI → V

63) Match the column-A and B about the embryonic development of human :-

	Column-A		Column-B
(i)	End of one month	(a)	Most of the major organ systems developed
(ii)	End of second month	(b)	Appearance of hair on the head
(iii)	End of 12 weeks	(c)	Heart formation
(iv)	During fifth month	(d)	Eye lid separated
(v)	End of about 24 weeks	(e)	Limbs and digits formation

	(i)	(ii)	(iii)	(iv)	(v)
(1)	c	e	a	b	d
(2)	c	a	e	d	b
(3)	c	c	b	a	d
(4)	d	e	b	a	c

(1) 1
(2) 2
(3) 3
(4) 4

64) Match the column

	Column-I		Column-II
(A)	Acrosin	(i)	Acrosomal reaction
(B)	Corpus luteum	(ii)	Morphogenetic movements
(C)	Gastrulation	(iii)	Progesterone
(D)	Capacitation	(iv)	Mammary gland
(E)	Colostrum	(v)	Sperm activation

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(1) A–iii, B–ii, C–v, D–iv, E-i

(2) A–i, B–ii, C–iii, D–iv, E–v

(3) A–i, B–iii, C–ii, D–v, E–iv

(4) A–v, B–ii, C–iv, D–i, E–iii

65) In a female undergoing tubectomy, which one of the following is correct :-

	Ovarian changes	Uterine Changes	Ovulation	Menstruation
1	No effect	No effect	Occur	Occur
2	No effect	No effect	Not occur	Stop
3	Stop	Stop	Not occur	Stop
4	No effect	Stop	Occur	Stop

(1) 1

(2) 2

(3) 3

(4) 4

66) Identify the correct match from the column I, II and III:-

Column-I	**Column-II**	**Column-III**
1. Labia majora	a. Two thin fold of skin	(i) Fourchette
2. Labia minora	b. Small cylindrical and erectile	(ii) Homologous to scrotum
3. Clitoris	c. Fleshy Folds of skin	(iii) Homologous to cowper's gland
4. Bartholin's gland	d. Greater vestibular gland	(iv) Homologous to penis

(1) 1-c-ii, 2-a-i, 3-b-iv, 4-d-iii

(2) 1-d-iv, 2-c-ii, 3-a-i, 4-b-iii

(3) 1-a-i, 2-b-ii, 3-c-iii, 4-d-iv

(4) 1-b-i, 2-a-ii, 3-d-iv, 4-c-iii

67) Which of the following is **mismatch** w.r.t. menstrual cycle?

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1	Menarche	First menstruation
2	Menstrual cycle	Cycle of events in between end of two successive menstruations
3	Menstrual phase	Occurrence of menstrual flow
4	Lack of menstruation	Can be indicative of pregnancy

(1) 1

(2) 2

(3) 3

(4) 4

68) Arrange the components of mammary gland. (from proximal to distal)
(a) Mammary duct
(b) Lactiferous duct
(c) Alveoli
(d) Mammary ampulla
(e) Mammary tubules
Choose the most appropriate answer from the options given below :

(1) (c) → (a) → (d) → (e) → (b)

(2) (b) → (c) → (e) → (d) → (a)

(3) (c) → (e) → (a) → (d) → (b)

(4) (e) → (c) → (d) → (b) → (a)

69)

Assertion – If both the ovaries are removed after the first Trimester of pregnancy, still there would be normal growth of foetus.

Reason– After first trimester placenta secretes sufficient progesterone.

(1) Both Assertion & Reason are true and the reason is the correct-explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true but-Reason is false

(4) Both are false statements

70) Which one of the following is not the function of placenta ?

(1) Facilitates removal of carbon dioxide and waste material from embryo.

(2) Secretes oxytocin during parturition.

(3) Facilitates supply of oxygen and nutrients to embryo.

(4) Secretes estrogen.

71) Which of following event will not occur in a female undergoing hysterectomy ?

(1) Formation of graffian follicle

(2) Ovulation

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(3) Menstruation

(4) Completion of meiosis-I

72) Fill in the blanks :-
(a) Zygote divides to form ...1... which is implanted to uterus.
(b) The structure which provides vascular connection between foetus and uterus is called ...2...
(c) Inner cell mass contain certain cells called ...3... which have the potency to give rise to all the tissues and organs.
(d) By the end of ...4..., most of the major organ systems are formed, for example, the limbs and external genital organs are well-developed.
(e) Immediately after implantation, the ...5... differentiates into an outer layer called ectoderm and an inner layer called endoderm.

(1) 1–morula, 2–umbilical cord, 3–trophoblast, 4–second trimester, 5–stem cells.

(2) 1–blastocyst, 2–placenta, 3–stem cells, 4–first trimester, 5–trophoblast.

(3) 1–blastocyst, 2–umbilical cord, 3–stem cells, 4–second trimesters, 5–inner cell mass.

(4) 1–blastocyst, 2–placenta, 3–stem cells, 4–first trimester, 5–inner cell mass.

73) Which of the following is largest in size ?

(1) Mature sperm cells.

(2) Mature egg cells.

(3) Polar body.

(4) They are all approximately the same size.

74) How many of following hormones are secreted by woman only during pregnancy?
hCG, hPL, Estrogen, Progesterone, Cortisol, Prolactin, Thyroxine.

(1) 1

(2) 2

(3) 3

(4) 4

75) Identify correct match of column I, II and III

	Column-I		**Column-II**		**Column-III**
A	Sertoli cells	a	H.C.G.	i	Promote Spermatogenesis
B	Leydig cells	b	Inhibin	ii	Myometrium hypertrophy
C	Trophoblast	c	Testosterone	iii	Suppression of Spermatogenesis
D	Ovary	d	Estrogen	iv	Maintain the corpus luteum

(1) A-b-iii, B-c-i, C-a-iv, D-d-ii

(2) A-c-ii, B-b-i, C-d-iii, D-c-iv

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(3) A-a-i, B-d-ii, C-b-iv, D-c-iii

(4) A-d-iv, B-c-iii, C-b-ii, D-a-i

76) The second maturation division of the mammalian ovum occurs :

(1) In the Graafian follicle following the first maturation division

(2) Shortly after ovulation before the ovum makes entry into the Fallopian tube

(3) Until after the ovum has been penetrated by a sperm

(4) Until the nucleus of the sperm has fused with that of the ovum

77) Find out incorrect statement about extraembryonic membrane:-

(1) Amnion - It is formed by the layer of amniogenic cells and visceral extraembryonic mesoderm.

(2) Chorion - It participate in the formation of placenta

(3) Yolk sac - It act as haemopoetic organ of foetus

(4) Allantois - In human, it store the excretory wastes

78)

Second polar body is released during oogenesis.:-

(1) At the time of fertilization

(2) After completion of meiosis II

(3) At the time of formation of mature haploid ovum

(4) All of the above

79)

The removal of both the testis before puberty results in the subject becomes

(1) Sterile

(2) Childish

(3) Both (1) & (2)

(4) Not any effect

80) Evaluate the following four statements based on the provided text and select the option that identifies the **correct** statements.
Statement I: A plasma membrane covers only the head and middle piece of the sperm, not the tail.
Statement II: Secretions from the epididymis, vas deferens, seminal vesicle, and prostate are necessary for the maturation and motility of sperms.
Statement III: For normal fertility, at least 60% of the total sperms ejaculated must exhibit vigorous motility.
Statement IV: The functions of the male sex accessory ducts and glands are maintained by **androgens** (testicular hormones).

(1) I, II, and III are correct.

(2) II and IV are correct.

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(3) I, III, and IV are correct.

(4) II, III, and IV are correct.

81) Which of the following contributes in the formation of seminal plasma.
(a) Sertoli cells
(b) Seminal vesicle
(c) Spermatogonia
(d) Leydig cells
(e) Bulbourethral gland
(f) Prostate gland

(1) b, c, e, f

(2) a, b, c, f

(3) b, c, d, e, f

(4) only b, e,, f

82) How many of the following are secondary sex organs in males ?

Uterus, Epididymis, Vas deferens, Prostate, Fallopian tube, Seminal vesicle, Penis, Vagina, Testes, Bulbourethral glands

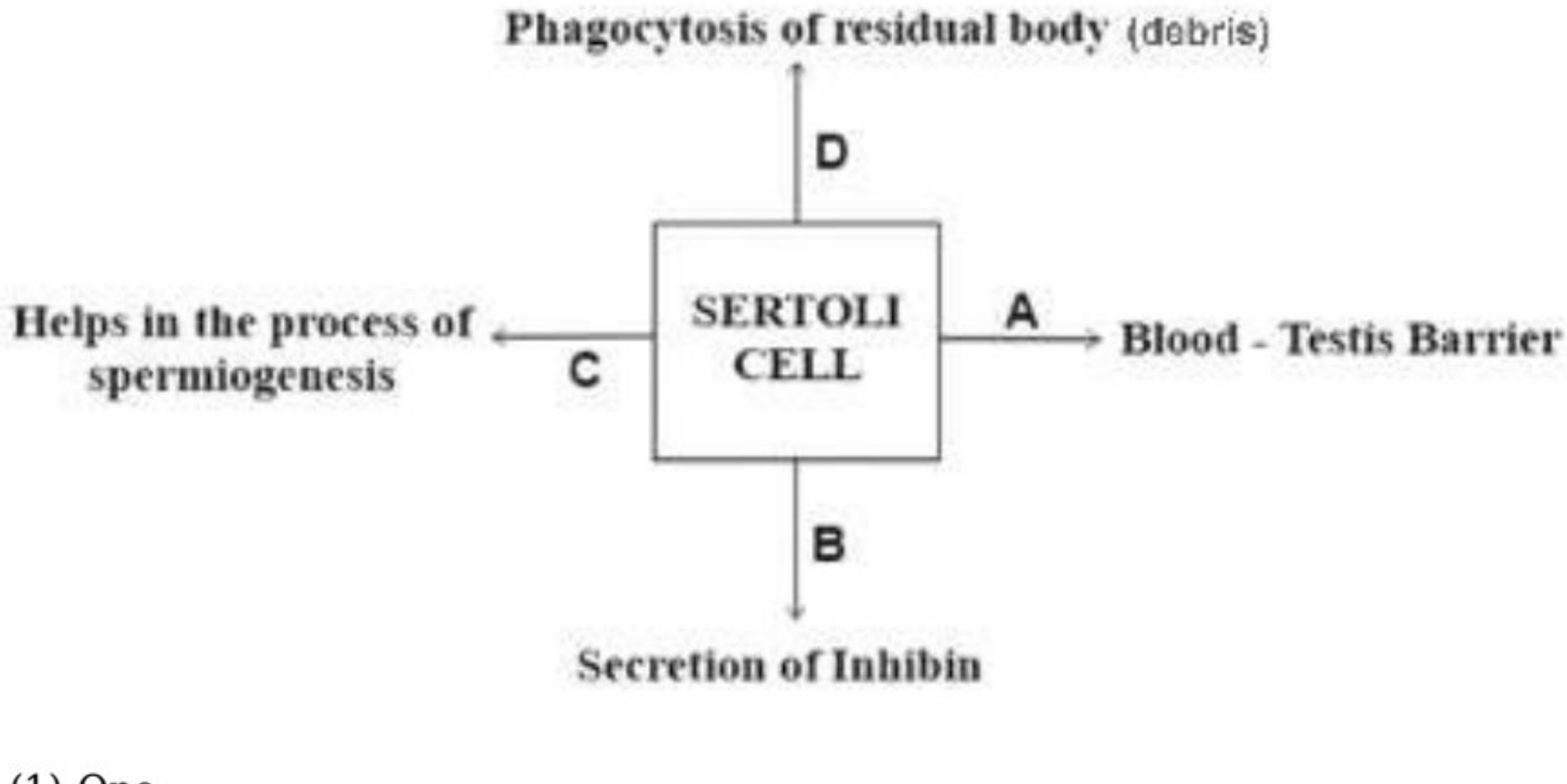
(1) Five

(2) Six

(3) Seven

(4) Four

83) How many connections are **correctly** established in the following:



(1) One

(2) Two

(3) Three

(4) All

84) How many of the following structures may found in fallopian tube ?
Morula, Secondary oocyte, Primary oocyte, Secondary follicle, Ovum, Zygote, Blastocyst, Gastrula

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(1) 3
(2) 2
(3) 4
(4) 5

85) Listed below are some of the components of the human female mammary gland:
(a) Mammary duct
(b) Lactiferous duct
(c) Alveoli
(d) Ampulla
Among these, choose the one that represents the most distal part.

(1) (a)
(2) (b)
(3) (c)
(4) (d)

86)

Find out correct match from the table :-

(i)	Ovulation	LH surge	Mid cycle
(ii)	Bleeding	With drawl of prostaglandin	1st day of M.C.
(iii)	Bleeding stops	Estrogen	5th day

(1) (ii) only
(2) (i) only
(3) (i) and (iii)
(4) (iii) only

87) Zygote is :-

(1) Diploid
(2) Vital link between two generation
(3) Universal structure in sexually reproducing organism
(4) All of the above

88)

A human female is born with a million of eggs (primary oocyte) at the time of birth, only some 500 eggs get a chance of maturity. What is the destiny of rest of the eggs?

(1) Rest of the eggs differentiate back to thecal and granulosa cells
(2) Rest of the eggs nurture the dominant follicular cell
(3) Rest of the eggs move out of the ovary and are destroyed by leucocytes
(4) Rest of the eggs break down and are absorbed i.e. degenerative follicular atresia

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89) In a female if there is no entry of sperm then how many structures in the given list will not be formed ?
1st polar body, 2nd polar body, Secondary oocyte, Primary oocyte.

(1) 2

(2) 1

(3) 4

(4) 3

90)
How many of the following structures are part of external genitalia of a female ?

Mons pubis, Labia majora, Clitoris, Vaginal canal, Uterus, Fallopian tube, Hymen.

(1) 5

(2) 4

(3) 3

(4) 2

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	3	2	3	1	1	1	2	4	2	4	2	2	2	3	2	3	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	3	2	3	4	1	4	3	1	4	2	3	1	2	1	3	1	4	3	2
Q.	41	42	43	44	45															
A.	4	1	1	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	3	1	1	4	4	1	4	3	4	1	2	1	1	1	4	3	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	3	2	4	2	2	1	1	3	3	2	2	4	4	2	4	4	3	2
Q.	86	87	88	89	90															
A.	3	1	1	1	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	4	2	2	4	1	4	4	2	2	3	4	4	1	4	2	1	4	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	1	1	4	3	2	3	2	1	2	4	4	3	3	2	2	4	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	3	4	3	2	2	2	2	2	4	2	3	1	4	3	2	4	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	2	1	3	1	1	2	3	1	2	3	4	2	2	1	3	4	4	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	4	3	2	3	4	4	2	2										

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SOLUTIONS

PHYSICS

1)

Conceptual

2)

Conceptual

19) **Question explanation:** Find the electrostatic potential of a charged soap bubble of radius r in equilibrium when internal and external air pressures are equal.

Concept: Equilibrium of a charged soap bubble where electrostatic pressure balances excess pressure due to surface tension.

Solution:

Here, $P_s = \frac{4T}{r}$

and $P_e = \frac{\sigma^2}{2\varepsilon_0}$

Since bubble is in equilibrium

$P_e = P_s$

$$\frac{\sigma^2}{2\varepsilon_0} = \frac{4T}{r} \Rightarrow \sigma = \sqrt{\frac{8T\varepsilon_0}{r}}$$

The potential V of a spherical conductor is given by $\frac{\sigma r}{\varepsilon_0}$

$$V = \frac{r}{\varepsilon_0} \times \sqrt{\frac{8T\varepsilon_0}{r}}$$

$$V = \sqrt{\frac{r^2}{\varepsilon_0^2} \times \frac{8T\varepsilon_0}{r}}$$

$$V = \sqrt{\frac{8Tr}{\varepsilon_0}}$$

Final answer: Option (2)

29) **Question Explanation:**
Field due to continuous charge.

Concept:
Electric Field Integration.

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Solution:

To find the field at point P, consider a small element da: at distance a from end A. The distance of this element from P is (2L – x).

The small electric field is $dE = \frac{K\,dq}{r^2} = \frac{K(\lambda dx)}{(2L \cdot x)^2}$

Substitute $\lambda = a(2L - x)^2$:

$dE = \frac{K\alpha(2L-x)^2}{(2L \cdot x)^2}\,dx = K\alpha\,dx.$

Integrating from x = 0 to x = L:

$E = \int_0^L K\alpha dx = K\alpha(x)_0^L = K\alpha L.$

Final Answer: Option (1)

44) **1. Question Explanation:**

A particle with charge –q and mass m orbits a fixed charge +Q. We need to find the relationship between the radius and the time period T of this circular motion.

2. Concept Based:

This problem deals with Coulomb's Law, centripetal force, and the motion of charged particles in an electrostatic field. The subtopic is Circular Motion of Charged Particles under Coulomb Force.

3. Formula Used:

Coulomb force: $F_{Coulomb} = \frac{kQq}{r^2}$

Centripetal force: $F_{centripetal} = \frac{mv^2}{r}$

Time period relation: $T = \frac{2\pi r}{v}$

4. Calculation:

From Coulomb's law, the force acting on the particle due to the electrostatic interaction is balanced by the centripetal force required for circular motion:

$$\frac{kQq}{r^2} = \frac{mv^2}{r}$$

Solving for v, we get:

$$v = \sqrt{\frac{kQq}{mr}}$$

Next, using the time period relation $T = \frac{2\pi r}{v}$, we substitute v to get:

$$T = \frac{2\pi r}{\sqrt{\frac{kQq}{mr}}} = 2\pi\sqrt{\frac{mr^3}{kQq}}$$

Squaring both sides:

$$T^2 = \frac{4\pi^2 mr^3}{kQq}$$

Thus, we obtain:

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$$r^3 = \frac{\frac{kQq}{4\pi^2 m}}{} T^2$$

Since $k = \frac{1}{4\pi\in_0}$, substituting k into the equation gives:

$$r^3 = \frac{\frac{Qq}{16\pi^3\in_0 m}}{} T^2$$

Hence, the correct option is Option 2.

CHEMISTRY

46) $V = 11.2 \times M$

$$M = \frac{V}{11.2} = \frac{15}{11.2} = 1.35\,M$$

$N = M \times Vf = 1.35 \times 2$

$= 2.66\ N$

48) **Question Explanation:**

We are given 9.45 g of oxalic acid in 250 mL of solution. We need to find how much water should be added to make the solution decinormal (N/10).

Concept:
Normality (N) = (Equivalent weight × Number of equivalents) / Volume (L)
For oxalic acid ($H_2C_2O_4 \cdot 2H_2O$), n-factor = 2, so
Equivalent weight = 126 / 2 = 63 g/equiv.

Solution:
Equivalents of solute = 9.45 / 63 = 0.15 equiv.

For N/10 (0.1 N) solution:
Volume (in L) = Equivalents / Normality = 0.15 / 0.1 = 1.5 L

Thus, the required final volume = 1.5 L = 1500 mL.
Present volume = 250 mL
Water to be added = 1500 – 250 = 1250 mL = 1.25 L
Final Answer: Option 3

49)

Question Explanation : Find Normality and molality of a 2.05 M H_2SO_4 solution with density 1.08 g mL^{-1}.
Concept : This question is based on Normality for diprotic acid = 2 × M.
Molality = moles solute per kg solvent.

Solution :
For 1.00 L solution:
Moles H_2SO_4 = 2.05 mol.
Mass of solution = 1.08 g mL^{-1} × 1000 mL = 1080 g.

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Molar mass $H_2SO_4 \approx 98.08$ g mol^{-1} → mass solute = 2.05 × 98.08 = 201.06 g.
Mass of solvent = 1080 − 201.06 = 878.94 g = 0.87894 kg.

Normality = 2 × 2.05 = 4.10 N.
Molality = 2.05 mol / 0.87894 kg = 2.33 m (rounded).
Final Answer : Option 1

50)

Explain the Question : We need to calculate the moles of N_2 dissolved in water using Henry's law.
Concept : This question is based on Henry's law : $P = k_H X_g$.
Solution :
Given :
$k_H = 1.0 \times 10^5$ atm
Mole fraction of N_2 in air = 0.8
Water moles = 10
Total pressure = 5 atm
Partial pressure of N_2 :
P_{N_2} = mole fraction of N_2 × total pressure = 0.8 × 5 = 4 atm
Henry's law :

$$X_{N_2} = \frac{P_{N_2}}{k_H} = \frac{4}{1.0 \times 10^5} = 4.0 \times 10^{-5}$$

Moles of N_2 dissolved :
$n = X_{N_2} \times n_{water} = 4.0 \times 10^{-5} \times 10 = 4.0 \times 10^{-4}$ moles
Final answer : 1

52)

Question Explanation: Total vapour pressure of a binary mixture is $P = 180\ X_A + 90$; find ratio of pure component vapour pressures $P°A : P°_B$.

Concept: For ideal solution, $P_{total} = X_A\ P°_A + X_B\ P°_B$; $X_B = 1 - X_A \rightarrow P = X_A\ P°_A + (1 - X_A)\ P°_B = X_A\ (P°_A - P°_B) + P°_B$. Compare with given equation to find $P°_A / P°_B$.

Solution:
Given: $P = 180\ X_A + 90$.
Compare with: $P = X_A\ (P°_A - P°_B) + P°_B$.
So, $P°_A - P°_B = 180 \rightarrow (1)$
$P°_B = 90 \rightarrow (2)$
From (1), $P°_A = 180 + 90 = 270$.
Ratio $P°_A : P°_B = 270 : 90 = 3 : 1$.

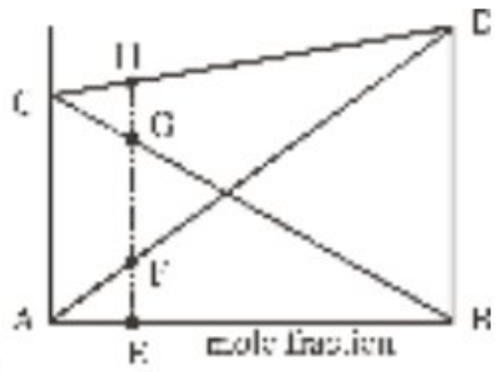
Final Answer: Option 4.

53)

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56) Mole fraction

Plots AD show vapour pressure of B containing a volatile component.

Plots BC show vapour pressure of A containing B volatile component.

Plot CD show vapour pressure of liquid solution containing A and B. Volatile components.

$P = P_A + P_B$

$\square$ $EH = EF + EG$

61) **Question Explanation:**

To find out molality of solution if RLVP is given

Concept:

RLVP

Solution:

$$\text{RLVP} = \frac{\Delta P}{P^o} = \frac{n}{n+N} = 0.0125$$

$$X_{solute} = \frac{1}{80} \quad X_{solvent} = \frac{79}{80}$$

$$\frac{X_{solute}}{X_{solvent}} = \frac{1}{79} = \frac{n_{solute}}{n_{solvent}}$$

$$m = \frac{1}{79} \times 1000 = 0.7$$

Final Answer: Option (1)

62) 1M

$HA \rightleftharpoons H^+ + A^-$

$(C - C\alpha) \quad C\alpha \quad C\alpha$

$[H^+] = 10^{-2}$ $\square$ $P^H = 2$

$C\alpha = 10^{-2}$

$1 \times \alpha = 10^{-2}$

$\alpha = 10^{-2}$

$\pi = i\,CRT$

$=(1 + \alpha)\,CRT$

$=(1 + 10^{-2}) \times 1RT$

$1.01\,RT$

63)

Question Explanation: Check correctness of assertion and direction of osmosis.

Concept:

- Osmosis: solvent flows from dilute → concentrated
- Reverse osmosis: requires external pressure > osmotic pressure

Solution:

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Assertion: Correct (pressure must exceed osmotic pressure)
Reason: Incorrect (wrong direction given)
A is correct, R is incorrect

Final Answer: Option (3)

65)

Conceptual

67) $\text{RLVP} = \frac{\text{in}}{\text{in} + \text{N}}$ (i = Vant Hoff factor)

$$0.167 = \frac{2n}{2n + 10}$$

$$\Rightarrow 0.334n + 1.67 = 2n$$

$$\Rightarrow n = \frac{1.67}{2 - 0.334} = \frac{1.67}{1.666} \simeq 1$$

69) molality (m) $= \frac{n_{solute}}{w_{solvent(kg)}}$

$$m = \frac{0.5}{500} \times 1000 = 1$$

$\Delta T_f = K_f \times i \times m$

$= 1.8 \times 3 \times 1 = 5.4$

$T_f = 0 - 5.4° = -5.4°C$

$\Delta T_b = K_b \times i \times m$

$= 0.5 \times 3 \times 1 = 1.5$

$T_b = 100 + 1.5 = 101.5°C$

71) Conceptual

74)

Explain the question : We are to determine the rate law for reaction A + 5B + 6C → 3L + 3M using given initial rate data.
Concept : This question is based on determination of order of reaction using the initial rate method.
Solution : Let rate = $k[A]^m[B]^n[C]^p$
From Exp. (1) & (2): [A] is halved (0.02 → 0.01), rate halves ($2.08\times10^{-3} \rightarrow 1.04\times10^{-3}$).
$\Rightarrow m = 1$
From Exp. (1) & (3): [B] doubles (0.02 → 0.04), rate doubles ($2.08\times10^{-3} \rightarrow 4.16\times10^{-3}$).
$\Rightarrow n = 1$
From Exp. (1) & (4): [C] doubles (0.02 → 0.04), rate becomes four times ($2.08\times10^{-3} \rightarrow 8.32\times10^{-3}$).
$\Rightarrow p = 2$
Hence, rate law = $k[A]^1[B]^1[C]^2$
Final Answer : 1

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76) (i) Rate law from slow step

$R = K(N_2O_2)[H_2]$

(ii) N_2O_2 is intermediate

$$K_{eq} = \frac{[N_2O_2]}{(NO)^2}$$

$[N_2O_2] = K_{eq}(NO)^2$

(iii) Put the value of N_2O_2 in rate law of slowest step

$R = K\,K_{eq}[NO]^2[H_2]$

Total order = 3

w.r.t. to H_2 = 1

77)

$$2NH_3 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} N_2 + 3H_2$$

$r_f = k_1 [NH_3]^2$

$r_b = k_2 [N_2] [H_2]^3$

$$r = -\frac{1}{2}\frac{dNH_2}{dt} = \frac{dN_2}{dt} = \frac{1}{3}\frac{dH_2}{dt}$$

from above

$r = r_f - r_b$

$$r = \frac{1}{3}\frac{dH_2}{dt}$$

$$\frac{dH_2}{dt} = 3k_1 [NH_3]^2 - 3k_2 [N_2] [H_2]^3$$

78)

Conceptual

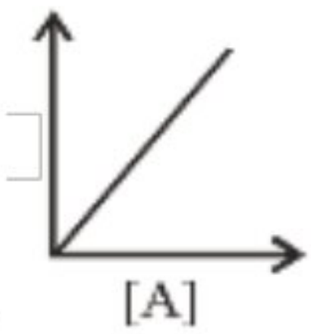
82) **Question Explanation :** We need to find order of reaction using given graphs.

Concept : Order of reaction.

Solution :

$nA \rightarrow B + C$

(i) Rate $\propto$ [A] $\Rightarrow$ order of reaction = 1



Rate

(ii) $y = -mx + C$

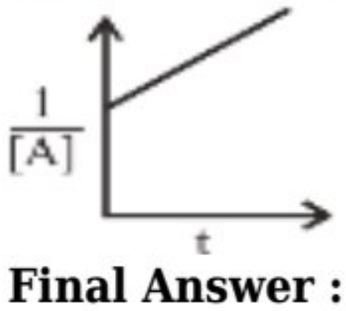
$[A]_t = -kt + [A]_0 \Rightarrow$ zero order reaction.

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(ii) $[A]^{t-1} = Kt + [A]_0^{-1} \Rightarrow$ order of reaction = 2



Final Answer : Option (4)

84)

For nth order reaction

$$K_t = \frac{1}{n-1}\left[\frac{1}{A_t^{n-1}} - \frac{1}{A_0^{n-1}}\right] \quad (n \neq 1)$$

$$Kt_{7/8} = \frac{1}{n-1}\left[\frac{1}{\left(\frac{A_0}{8}\right)^{n-1}} - \frac{1}{A_0^{n-1}}\right] \quad \ldots\ldots\ldots (1)$$

$$Kt_{1/2} = \frac{1}{n-1}\left[\frac{1}{\left(\frac{A_0}{2}\right)^{n-1}} - \frac{1}{A_0^{n-1}}\right] \quad \ldots\ldots\ldots (2)$$

Reaction (1) / Reaction (2)

$$\boxed{\frac{t_{0.875}}{t_{0.50}} = \frac{t_{7/8}}{t_{1/2}} = \frac{8^{n-1}-1}{2^{n-1}-1}}$$

85) $\quad 2A \rightarrow B$

t = 0 $\quad$ a $\quad$ 0

t = t $\quad$ a-2x $\quad$ x

at point 'P'

a–2x = x

x = a/3

$\therefore \quad 2x = 2a/3$

$\therefore \quad t = t_{2/3}$

86)

Question Explanation: We are asked to find the halt-life of a second order reaction given that it is 40% complete in 30 minutes.

Concept: Second-Order Kinetics

$$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

Halt life for second order reaction:

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$$t_{1/2} = \frac{1}{k[A]_0}$$

Solution:

- Let initial concentration = $[A]_0$
- After 40% completion, $[A] = 0.6[A]_0$
- Using integrated law:

$$\frac{1}{0.6[A]_0} - \frac{1}{[A]_0} = k.30$$

$$\frac{1}{[A]_0}\left(\frac{1}{0.6} - 1\right) = k.30$$

$$\frac{1}{[A]_0}(0.6667) = k.30$$

$$k = \frac{0.6667}{30[A]_0} = \frac{0.02222}{[A]_0}$$

- Half life:

$$t_{1/2} = \frac{1}{k[A]_0} = \frac{1}{0.02222} \approx 45\,\text{min}$$

Final Answer: (3)

88)

$$2.3 \times 10^{-5}\ s^{-1} \times 90 \times 60s = 2.303 \log\left(\frac{A_0}{A_t}\right)$$

$$\log\left(\frac{A_0}{A_t}\right) = 0.054 \Rightarrow [A_t = 0.885\ A^0]$$

89) Conceptual

BIOLOGY

97) NCERT-XII, Pg. # 31

102)

Question Explanation: The correct description of the vegetative and generative cells in a mature pollen grain.

Concept: Pollen Grain Structure

Solution: The vegetative cell is the larger cell with abundant food reserves. The A generative cell is the smaller, spindle-shaped cell that floats in the cytoplasm of the vegetative cell.

Final Answer: The generative cell is spindle-shaped and floats in the cytoplasm of the vegetative cell, which is larger with abundant food reserve.

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103)

Question Explanation: Question is asking ,given two statements about the structure of seed,select correct or incorrect statement

Concept: Seed

Solution:
Statement I: In angiosperms, the seed is often described as a fertilised ovule and is formed outside the fruits.-Incorrect
Correct is -In angiosperms, the seed is the final product of sexual reproduction. It is often described as a fertilised ovule. Seeds are formed inside fruits.
Statement II: The cotyledons of the embryo are simple structures, generally thick and swollen due to storage of food reserves.-Correct

Final answer: 4

104)

Question Explanation :

Evaluate assertion and reason about pollination in Vallisneria.

Concept :

Hydrophily

Solution :

Assertion – It says short stalk female flowers reach surface → Incorrect

Actual: female flowers have long coiled stalk to reach the surface .

Reason – Male flowers/pollen are released inside water surface → Incorrect (they are released and float on the surface)

Final Answer : Option 4

105)

Question Explanation: The structure whose development precedes that of the embryo in angiosperms.

Concept: Embryo and Endosperm Development

Solution: The formation of the nutritive tissue, the endosperm, always begins and is partly completed before the zygote starts to divide, ensuring assured nutrition for the developing embryo**.**

Final Answer: Embryo

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108) NCERT Pg. # 31(E) & 33(H)

109) NCERT (XII) Page # 29

110) NCERT Page # 22

114) NCERT XII Pg.# 34(E), 36(H)

115) NCERT-XII, Pg.# 13

117) NCERT-XII, Page No. : 05,08

118) NCERT Page (27-E), (27-H)

119)

Question Explanation : The question asks to identify evaluate statements about the allergenic and nutritional uses of pollen grains.

Concept : Pollen Allergy and Uses

Solution:

Statement (I) is correct: Pollen grains from certain plants, like Parthenium (carrot grass), do cause severe allergies and bronchial afflictions such as asthma in many people.

Statement (II) is also correct: Pollen grains are indeed rich in nutrients, including carbohydrates, proteins, and vitamins, and are widely used as dietary supplements in the form of tablets and syrups, often marketed for enhancing athletic performance.

Final Answer : option (2) Both Statement - I & II are correct.

121) NCERT-XII Pg#20

125) NCERT Pg. No. # 29

126) NCERT-XII Pg. # 31

128)

Question Explanation : The question asks to identify the group of plants whose seeds contain a coleoptile and a coleorhiza.

Concept : This question is based on Monocot Embryo Structure.

Solution :

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The coleoptile and coleorhiza are protective sheaths that cover the embryonic shoot (plumule) and embryonic root (radicle), respectively. These two structures are a characteristic feature of the embryos of monocotyledonous plants, especially those belonging to the Grass family (Poaceae), such as maize, wheat, and rice. Dicot seeds lack these sheaths.

Final Answer : option (4) Grass family

129) NCERT (XIIth) Pg. # 25

133) NCERT Page No.

134) NCERT Pg # 24

135) NCERT Page No. 35

157) NCERT Page: 33, 34, 35

164)

Question explanation: Identify hormones that are secreted **only during pregnancy** in women.
Concept: Human Reproduction – Pregnancy Hormones
Solution:

A. **hCG (human chorionic gonadotropin):** Secreted **only during pregnancy** by placenta

B. **hPL (human placental lactogen):** Secreted **only during pregnancy** by placenta

C. **Estrogen:** Also secreted by ovaries normally

D. **Progesterone:** Also secreted by ovaries normally

E. **Cortisol:** Secreted by adrenal cortex in all individuals

F. **Prolactin:** Secreted normally for lactation, even in non-pregnant women

G. **Thyroxine:** Secreted normally by thyroid gland

Number of hormones secreted only during pregnancy: 2 → hCG, hPL

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Final answer: Option 2

165)

Conceptual

168)

NCERT Pg.# 49

170) **Explanation:** Statement I is Incorrect: "A plasma membrane envelops the whole body of sperm," which includes the head, neck, middle piece, and tail.
Statement III is Incorrect: "at least 60 per cent sperms must have normal shape and size and at least 40 per cent of them must show vigorous motility." Statement III incorrectly assigns the 60% threshold to vigorous motility (it should be 40%).
Final Answer : (2)

174)

Question Explanation – The question asks how many of the listed structures can be found in the fallopian tube during reproduction.

Concept – Events of fertilization and early development

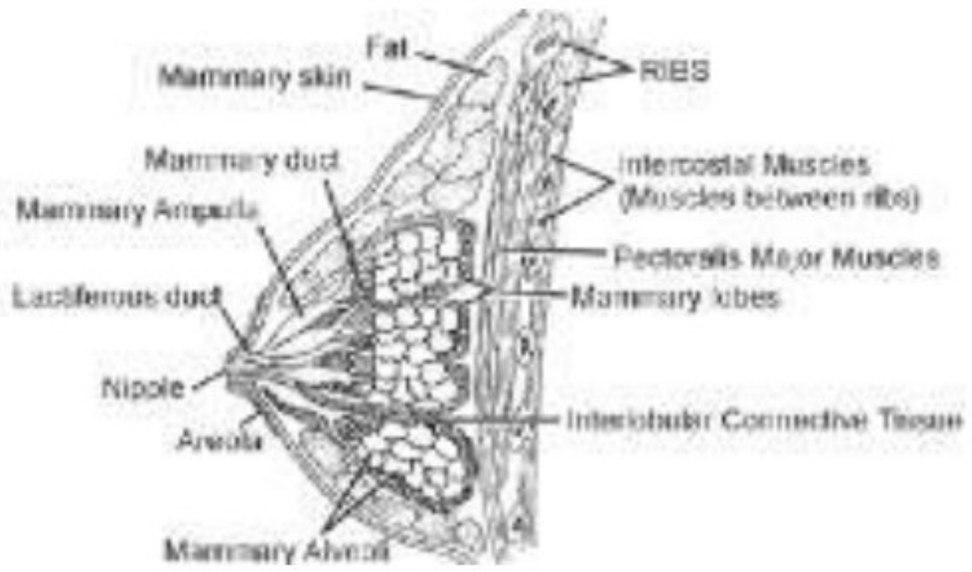
Solution –Present in fallopian tube:

Secondary oocyte → released during ovulation
Ovum → formed after fertilization
Zygote → formed in the ampulla region
Morula → early cleavage stage while moving toward the uterus

Thus, 4 structures are found in the fallopian tube.

Final Answer – Option [3]

175)



179)

Question Explanation : The question asks to count how many structures from the given list

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will not be formed if there is no entry of sperm.
Concept : Oogenesis
Solution : The correct answer is option **2**.

A. The **primary oocyte**, **secondary oocyte**, and **first polar body** are all formed before fertilization occurs.
B. The **second polar body**, however, is formed only after the secondary oocyte completes its second meiotic division, which is triggered by the entry of a sperm.
C. Therefore, only one structure—the second polar body—will not be formed if there is no sperm entry.

Final Answer : Option (2)

180) **Explain Question:**
Count how many given structures belong to female external genitalia.

Concept:
This question is based on female external genitalia.

Solution:
Total = 4 (Mons pubis, Labia majora, Clitoris, Hymen).

Final Answer: 2 (4)

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